

H0 — decision summary (1/7)

llama31-8b (ctx 4,096, 1,024 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.8771 p99 0.2416 median 0.003222 (132/1024 heads >= 0.05)
heads above 0.001:688 0.005:454 0.01:343 0.05:132 0.1:55 0.25:10 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 2.743 b cont 2.867 b delta -0.148 b 24.4% of heads niah>cont
HEADS IN BAND 46.3% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.93x (geometric mean of max(gain,1) over all heads)
in-band heads median 3.0x p90 4.6x
ladder width median 2.81 b IQR [2.46, 3.14] 100% of heads above 1.4
tau median 1.94 excl. sinks 1.92
top-1 weight median 0.6534 95% mass in 97 tokens (2.6% of ctx)
B=2b vs uniform 37.5x vs eviction 2.0x vs BEST corner 2.0x evicts 61%
B=3b vs uniform 23.5x vs eviction 1.8x vs BEST corner 1.8x evicts 45%
quantizer alpha @2b 0.941 (1.0 = no temperature distortion); spearman@2b 0.743 [H2 answered]
heads where eviction beats a 1-bit tier: 99%
heads where eviction beats a 2-bit tier: 14%
linearization check (predicted/measured gain @3b): 1.23 (1.0 = exact)
VERDICT: GO — 46% of heads are in band (1.93x routed). The interior owns a substantial share of the model. Proceed to the nested-coding test.

llama31-8b (ctx 8,192, 1,024 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.6047 p99 0.1748 median 0.001845 (79/1024 heads >= 0.05)
heads above 0.001:593 0.005:383 0.01:277 0.05:79 0.1:28 0.25:7 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 2.896 b cont 3.116 b delta -0.231 b 19.1% of heads niah>cont
HEADS IN BAND 39.4% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.78x (geometric mean of max(gain,1) over all heads)
in-band heads median 2.9x p90 4.5x
ladder width median 3.00 b IQR [2.62, 3.37] 100% of heads above 1.4
tau median 2.07 excl. sinks 2.06
top-1 weight median 0.6464 95% mass in 137 tokens (1.8% of ctx)
B=2b vs uniform 43.1x vs eviction 1.9x vs BEST corner 1.9x evicts 62%
B=3b vs uniform 25.7x vs eviction 1.7x vs BEST corner 1.6x evicts 47%
quantizer alpha @2b 0.943 (1.0 = no temperature distortion); spearman@2b 0.722 [H2 answered]
heads where eviction beats a 1-bit tier: 99%
heads where eviction beats a 2-bit tier: 19%
linearization check (predicted/measured gain @3b): 1.13 (1.0 = exact)
VERDICT: GO — 39% of heads are in band (1.78x routed). The interior owns a substantial share of the model. Proceed to the nested-coding test.

llama31-8b (ctx 16,384, 1,024 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.7400 p99 0.1670 median 0.001497 (67/1024 heads >= 0.05)
heads above 0.001:561 0.005:369 0.01:264 0.05:67 0.1:26 0.25:7 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 3.059 b cont 3.458 b delta -0.345 b 12.0% of heads niah>cont
HEADS IN BAND 33.0% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.65x (geometric mean of max(gain,1) over all heads)
in-band heads median 3.0x p90 4.4x
ladder width median 3.21 b IQR [2.79, 3.59] 100% of heads above 1.4
tau median 2.22 excl. sinks 2.21
top-1 weight median 0.6377 95% mass in 229 tokens (1.5% of ctx)
B=2b vs uniform 48.3x vs eviction 1.8x vs BEST corner 1.8x evicts 63%
B=3b vs uniform 27.3x vs eviction 1.4x vs BEST corner 1.4x evicts 48%
quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.672 [H2 answered]
heads where eviction beats a 1-bit tier: 99%

H0 — decision summary (2/7)

llama31-8b (ctx 32,768, 1,024 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.8017 p99 0.1458 median 0.000826 (61/1024 heads >= 0.05)
heads above 0.001:492 0.005:321 0.01:236 0.05:61 0.1:22 0.25:6 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 3.328 b cont 3.766 b delta -0.418 b 9.1% of heads niah>cont
HEADS IN BAND 25.6% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.49x (geometric mean of max(gain,1) over all heads)
in-band heads median 2.9x p90 4.7x
ladder width median 3.49 b IQR [3.07, 4.03] 100% of heads above 1.4
tau median 2.41 excl. sinks 2.41
top-1 weight median 0.6249 95% mass in 264 tokens (0.9% of ctx)
B=2b vs uniform 56.4x vs eviction 1.6x vs BEST corner 1.5x evicts 64%
B=3b vs uniform 29.7x vs eviction 1.2x vs BEST corner 1.2x evicts 49%
quantizer alpha @2b 0.943 (1.0 = no temperature distortion); spearman@2b 0.642 [H2 answered]
heads where eviction beats a 1-bit tier: 99%
heads where eviction beats a 2-bit tier: 26%
linearization check (predicted/measured gain @3b): 0.98 (1.0 = exact)
VERDICT: NARROW – 26% of heads are in band (1.49x routed). Real but partial. Ship the router as part of the method and make the regime boundary a contribution.

llama31-8b (ctx 65,536, 1,024 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.7196 p99 0.1534 median 0.000277 (44/1024 heads >= 0.05)
heads above 0.001:406 0.005:259 0.01:175 0.05:44 0.1:23 0.25:5 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 3.643 b cont 4.259 b delta -0.554 b 5.5% of heads niah>cont
HEADS IN BAND 17.5% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.33x (geometric mean of max(gain,1) over all heads)
in-band heads median 2.9x p90 4.2x
ladder width median 3.84 b IQR [3.44, 4.37] 100% of heads above 1.4
tau median 2.67 excl. sinks 2.67
top-1 weight median 0.6057 95% mass in 347 tokens (0.6% of ctx)
B=2b vs uniform 62.9x vs eviction 1.3x vs BEST corner 1.3x evicts 65%
B=3b vs uniform 32.4x vs eviction 1.1x vs BEST corner 1.1x evicts 51%
quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.632 [H2 answered]
heads where eviction beats a 1-bit tier: 99%
heads where eviction beats a 2-bit tier: 30%
linearization check (predicted/measured gain @3b): 0.90 (1.0 = exact)
VERDICT: NARROW – 17% of heads are in band (1.33x routed). Real but partial. Ship the router as part of the method and make the regime boundary a contribution.

llama31-8b (ctx 131,072, 1,024 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.8572 p99 0.2670 median 0.000205 (57/1024 heads >= 0.05)
heads above 0.001:363 0.005:239 0.01:174 0.05:57 0.1:30 0.25:11 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 4.110 b cont 4.448 b delta -0.352 b 14.8% of heads niah>cont
HEADS IN BAND 12.7% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.25x (geometric mean of max(gain,1) over all heads)
in-band heads median 2.8x p90 4.3x
ladder width median 4.27 b IQR [3.77, 4.74] 100% of heads above 1.4
tau median 2.96 excl. sinks 2.96
top-1 weight median 0.4068 95% mass in 1158 tokens (1.0% of ctx)
B=2b vs uniform 68.4x vs eviction 1.2x vs BEST corner 1.2x evicts 67%
B=3b vs uniform 33.5x vs eviction 1.0x vs BEST corner 1.0x evicts 52%
quantizer alpha @2b 0.943 (1.0 = no temperature distortion); spearman@2b 0.612 [H2 answered]
heads where eviction beats a 1-bit tier: 99%

H0 — decision summary (3/7)

llama33-70b (ctx 8,192, 5,120 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 4/4 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.3040 p99 0.0939 median 0.000544 (134/5120 heads >= 0.05)
heads above 0.001:2023 0.005:1035 0.01:688 0.05:134 0.1:47 0.25:3 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 2.155 b cont 2.312 b delta -0.149 b 18.4% of heads niah>cont
HEADS IN BAND 79.1% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 3.06x (geometric mean of max(gain,1) over all heads)
in-band heads median 3.6x p90 6.0x
ladder width median 2.21 b IQR [2.00, 2.51] 98% of heads above 1.4
tau median 1.51 excl. sinks 1.49
top-1 weight median 0.6973 95% mass in 32 tokens (0.4% of ctx)
B=2b vs uniform 31.7x vs eviction 2.6x vs BEST corner 2.5x evicts 57%
B=3b vs uniform 21.4x vs eviction 3.3x vs BEST corner 3.2x evicts 40%
quantizer alpha @2b 0.941 (1.0 = no temperature distortion); spearman@2b 0.767 [H2 answered]
heads where eviction beats a 1-bit tier: 97%
heads where eviction beats a 2-bit tier: 6%
linearization check (predicted/measured gain @3b): 1.35 (1.0 = exact)
VERDICT: GO — 79% of heads are in band (3.06x routed). The interior owns a substantial share of the model. Proceed to the nested-coding test.

llama33-70b (ctx 32,768, 5,120 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 4/4 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.5117 p99 0.0874 median 0.000121 (95/5120 heads >= 0.05)
heads above 0.001:1269 0.005:619 0.01:399 0.05:95 0.1:47 0.25:8 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 2.643 b cont 3.104 b delta -0.425 b 4.7% of heads niah>cont
HEADS IN BAND 58.5% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 2.29x (geometric mean of max(gain,1) over all heads)
in-band heads median 3.2x p90 5.4x
ladder width median 2.78 b IQR [2.52, 3.18] 99% of heads above 1.4
tau median 1.91 excl. sinks 1.90
top-1 weight median 0.7277 95% mass in 53 tokens (0.2% of ctx)
B=2b vs uniform 55.5x vs eviction 2.4x vs BEST corner 2.4x evicts 61%
B=3b vs uniform 30.4x vs eviction 2.4x vs BEST corner 2.3x evicts 45%
quantizer alpha @2b 0.943 (1.0 = no temperature distortion); spearman@2b 0.675 [H2 answered]
heads where eviction beats a 1-bit tier: 98%
heads where eviction beats a 2-bit tier: 9%
linearization check (predicted/measured gain @3b): 0.96 (1.0 = exact)
VERDICT: GO — 58% of heads are in band (2.29x routed). The interior owns a substantial share of the model. Proceed to the nested-coding test.

llama33-70b (ctx 131,072, 5,120 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 3/4 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.6931 p99 0.0478 median 0.000042 (51/5120 heads >= 0.05)
heads above 0.001:583 0.005:236 0.01:146 0.05:51 0.1:27 0.25:8 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 4.713 b cont 5.223 b delta -0.379 b 12.3% of heads niah>cont
HEADS IN BAND 18.0% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.34x (geometric mean of max(gain,1) over all heads)
in-band heads median 3.0x p90 5.1x
ladder width median 4.84 b IQR [3.82, 5.78] 99% of heads above 1.4
tau median 3.34 excl. sinks 3.34
top-1 weight median 0.1495 95% mass in 4808 tokens (4.0% of ctx)
B=2b vs uniform 76.7x vs eviction 1.4x vs BEST corner 1.4x evicts 68%
B=3b vs uniform 36.9x vs eviction 1.0x vs BEST corner 1.0x evicts 54%
quantizer alpha @2b 0.943 (1.0 = no temperature distortion); spearman@2b 0.552 [H2 answered]
heads where eviction beats a 1-bit tier: 99%

H0 — decision summary (4/7)

mistral-7b (ctx 8,192, 1,024 heads)

INPUT VALIDITY PASS haystack real

task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]

head-level needle mass: max 0.9435 p99 0.7437 median 0.003191 (193/1024 heads >= 0.05)

heads above 0.001:683 0.005:465 0.01:381 0.05:193 0.1:129 0.25:67 [ADVISORY - MIN_MASS uncalibrated, not enforced]

RETIRED median-ladder gate (not enforced): niah 2.757 b cont 2.993 b delta -0.200 b 23.4% of heads niah>cont

HEADS IN BAND 51.3% (gain over best corner >= 2.0x at 3 b/token)

ROUTED GAIN 2.07x (geometric mean of max(gain,1) over all heads)

in-band heads median 3.2x p90 5.0x

ladder width median 2.87 b IQR [2.51, 3.30] 98% of heads above 1.4

tau median 1.98 excl. sinks 1.97

top-1 weight median 0.4411 95% mass in 387 tokens (5.1% of ctx)

B=2b vs uniform 34.2x vs eviction 2.1x vs BEST corner 2.1x evicts 61%

B=3b vs uniform 23.2x vs eviction 2.1x vs BEST corner 2.0x evicts 45%

quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.714 [H2 answered]

heads where eviction beats a 1-bit tier: 97%

heads where eviction beats a 2-bit tier: 9%

linearization check (predicted/measured gain @3b): 1.25 (1.0 = exact)

VERDICT: GO — 51% of heads are in band (2.07x routed). The interior owns a substantial share of the model. Proceed to the nested-coding test.

mistral-7b (ctx 32,768, 1,024 heads)

INPUT VALIDITY PASS haystack real

task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]

head-level needle mass: max 0.9474 p99 0.6913 median 0.001411 (141/1024 heads >= 0.05)

heads above 0.001:568 0.005:367 0.01:287 0.05:141 0.1:100 0.25:44 [ADVISORY - MIN_MASS uncalibrated, not enforced]

RETIRED median-ladder gate (not enforced): niah 3.050 b cont 3.385 b delta -0.296 b 19.5% of heads niah>cont

HEADS IN BAND 41.2% (gain over best corner >= 2.0x at 3 b/token)

ROUTED GAIN 1.82x (geometric mean of max(gain,1) over all heads)

in-band heads median 3.0x p90 4.5x

ladder width median 3.22 b IQR [2.79, 3.69] 98% of heads above 1.4

tau median 2.21 excl. sinks 2.20

top-1 weight median 0.4158 95% mass in 1392 tokens (4.6% of ctx)

B=2b vs uniform 40.7x vs eviction 2.0x vs BEST corner 2.0x evicts 63%

B=3b vs uniform 25.4x vs eviction 1.8x vs BEST corner 1.7x evicts 48%

quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.619 [H2 answered]

heads where eviction beats a 1-bit tier: 98%

heads where eviction beats a 2-bit tier: 20%

linearization check (predicted/measured gain @3b): 1.15 (1.0 = exact)

VERDICT: GO — 41% of heads are in band (1.82x routed). The interior owns a substantial share of the model. Proceed to the nested-coding test.

qwen15-moe-a2.7b (ctx 8,192, 384 heads)

INPUT VALIDITY PASS haystack real

task-level needle retrieved in 0/6 niah prompts (FAIL, need 50%) [measurement]

head-level needle mass: max 0.8715 p99 0.6935 median 0.002014 (67/384 heads >= 0.05)

heads above 0.001:221 0.005:155 0.01:133 0.05:67 0.1:42 0.25:22 [ADVISORY - MIN_MASS uncalibrated, not enforced]

RETIRED median-ladder gate (not enforced): niah 3.397 b cont 3.590 b delta -0.115 b 28.4% of heads niah>cont

HEADS IN BAND 29.4% (gain over best corner >= 2.0x at 3 b/token)

ROUTED GAIN 1.59x (geometric mean of max(gain,1) over all heads)

in-band heads median 3.2x p90 5.6x

ladder width median 3.47 b IQR [2.99, 3.96] 98% of heads above 1.4

tau median 2.39 excl. sinks 2.37

top-1 weight median 0.3786 95% mass in 173 tokens (2.3% of ctx)

B=2b vs uniform 44.4x vs eviction 1.7x vs BEST corner 1.7x evicts 64%

B=3b vs uniform 26.2x vs eviction 1.3x vs BEST corner 1.3x evicts 49%

quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.725 [H2 answered]

heads where eviction beats a 1-bit tier: 95%

H0 — decision summary (5/7)

qwen15-moe-a2.7b (ctx 32,768, 384 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 2/6 niah prompts (FAIL, need 50%) [measurement]
head-level needle mass: max 0.9243 p99 0.3916 median 0.000189 (34/384 heads >= 0.05)
heads above 0.001:147 0.005:107 0.01:81 0.05:34 0.1:23 0.25:9 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 4.295 b cont 4.755 b delta -0.268 b 16.7% of heads niah>cont
HEADS IN BAND 20.1% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.39x (geometric mean of max(gain,1) over all heads)
in-band heads median 3.6x p90 5.7x
ladder width median 4.42 b IQR [3.56, 5.47] 99% of heads above 1.4
tau median 3.06 excl. sinks 3.05
top-1 weight median 0.3335 95% mass in 600 tokens (2.0% of ctx)
B=2b vs uniform 57.1x vs eviction 1.2x vs BEST corner 1.2x evicts 66%
B=3b vs uniform 30.5x vs eviction 1.1x vs BEST corner 1.0x evicts 52%
quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.632 [H2 answered]
heads where eviction beats a 1-bit tier: 96%
heads where eviction beats a 2-bit tier: 36%
linearization check (predicted/measured gain @3b): 0.96 (1.0 = exact)
VERDICT: NARROW – 20% of heads are in band (1.39x routed). Real but partial. Ship the router as part of the method and make the regime boundary a contribution.

qwen3-30b-a3b-2507 (ctx 8,192, 1,536 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 2/4 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.7960 p99 0.4925 median 0.003895 (297/1536 heads >= 0.05)
heads above 0.001:1019 0.005:715 0.01:568 0.05:297 0.1:175 0.25:67 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 3.201 b cont 3.279 b delta -0.046 b 43.6% of heads niah>cont
HEADS IN BAND 10.2% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.25x (geometric mean of max(gain,1) over all heads)
in-band heads median 2.5x p90 3.3x
ladder width median 3.21 b IQR [2.84, 3.53] 100% of heads above 1.4
tau median 2.21 excl. sinks 2.20
top-1 weight median 0.5114 95% mass in 79 tokens (1.0% of ctx)
B=2b vs uniform 44.9x vs eviction 1.2x vs BEST corner 1.2x evicts 66%
B=3b vs uniform 26.0x vs eviction 1.1x vs BEST corner 1.1x evicts 51%
quantizer alpha @2b 0.943 (1.0 = no temperature distortion); spearman@2b 0.662 [H2 answered]
heads where eviction beats a 1-bit tier: 100%
heads where eviction beats a 2-bit tier: 61%
linearization check (predicted/measured gain @3b): 1.10 (1.0 = exact)
VERDICT: STOP – Only 10% of heads sit in the productive band and routing buys 1.25x overall. Uniform quantization or eviction already captures the gain almost everywhere; SIEVE would be a reframing without a result.

qwen3-30b-a3b-2507 (ctx 16,384, 1,536 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 2/4 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.8819 p99 0.5419 median 0.001979 (248/1536 heads >= 0.05)
heads above 0.001:883 0.005:608 0.01:502 0.05:248 0.1:156 0.25:58 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 3.312 b cont 3.500 b delta -0.166 b 30.8% of heads niah>cont
HEADS IN BAND 7.8% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.21x (geometric mean of max(gain,1) over all heads)
in-band heads median 2.5x p90 3.1x
ladder width median 3.38 b IQR [3.01, 3.79] 100% of heads above 1.4
tau median 2.33 excl. sinks 2.32
top-1 weight median 0.4825 95% mass in 160 tokens (1.1% of ctx)
B=2b vs uniform 46.5x vs eviction 1.1x vs BEST corner 1.1x evicts 66%
B=3b vs uniform 27.0x vs eviction 1.0x vs BEST corner 1.0x evicts 51%
quantizer alpha @2b 0.944 (1.0 = no temperature distortion); spearman@2b 0.605 [H2 answered]
heads where eviction beats a 1-bit tier: 100%

H0 — decision summary (6/7)

qwen3-30b-a3b-2507 (ctx 32,768, 1,536 heads)

INPUT VALIDITY PASS haystack real

task-level needle retrieved in 0/4 niah prompts (FAIL, need 50%) [measurement]

head-level needle mass: max 0.8618 p99 0.4405 median 0.000840 (215/1536 heads >= 0.05)

heads above 0.001:745 0.005:498 0.01:411 0.05:215 0.1:118 0.25:41 [ADVISORY - MIN_MASS uncalibrated, not enforced]

RETIRED median-ladder gate (not enforced): niah 3.432 b cont 3.579 b delta -0.100 b 34.4% of heads niah>cont

HEADS IN BAND 6.4% (gain over best corner >= 2.0x at 3 b/token)

ROUTED GAIN 1.17x (geometric mean of max(gain,1) over all heads)

in-band heads median 2.4x p90 3.1x

ladder width median 3.48 b IQR [3.13, 3.85] 100% of heads above 1.4

tau median 2.40 excl. sinks 2.40

top-1 weight median 0.4795 95% mass in 231 tokens (0.8% of ctx)

B=2b vs uniform 48.7x vs eviction 1.1x vs BEST corner 1.1x evicts 67%

B=3b vs uniform 27.4x vs eviction 1.0x vs BEST corner 1.0x evicts 52%

quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.566 [H2 answered]

heads where eviction beats a 1-bit tier: 100%

heads where eviction beats a 2-bit tier: 66%

linearization check (predicted/measured gain @3b): 1.05 (1.0 = exact)

VERDICT: STOP – Only 6% of heads sit in the productive band and routing buys 1.17x overall. Uniform quantization or eviction already captures the gain almost everywhere; SIEVE would be a reframing without a result.

qwen3-30b-a3b-2507 (ctx 131,072, 1,536 heads)

INPUT VALIDITY PASS haystack real

task-level needle retrieved in 2/4 niah prompts (PASS, need 50%) [measurement]

head-level needle mass: max 0.9289 p99 0.5178 median 0.000387 (188/1536 heads >= 0.05)

heads above 0.001:647 0.005:455 0.01:375 0.05:188 0.1:114 0.25:52 [ADVISORY - MIN_MASS uncalibrated, not enforced]

RETIRED median-ladder gate (not enforced): niah 4.119 b cont 4.572 b delta -0.382 b 16.9% of heads niah>cont

HEADS IN BAND 4.1% (gain over best corner >= 2.0x at 3 b/token)

ROUTED GAIN 1.11x (geometric mean of max(gain,1) over all heads)

in-band heads median 2.3x p90 3.0x

ladder width median 4.20 b IQR [3.66, 5.25] 100% of heads above 1.4

tau median 2.90 excl. sinks 2.90

top-1 weight median 0.4632 95% mass in 626 tokens (0.5% of ctx)

B=2b vs uniform 53.4x vs eviction 1.0x vs BEST corner 1.0x evicts 68%

B=3b vs uniform 29.2x vs eviction 1.0x vs BEST corner 1.0x evicts 54%

quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.521 [H2 answered]

heads where eviction beats a 1-bit tier: 100%

heads where eviction beats a 2-bit tier: 73%

linearization check (predicted/measured gain @3b): 0.99 (1.0 = exact)

VERDICT: STOP – Only 4% of heads sit in the productive band and routing buys 1.11x overall. Uniform quantization or eviction already captures the gain almost everywhere; SIEVE would be a reframing without a result.

qwen3-8b (ctx 8,192, 1,152 heads)

INPUT VALIDITY PASS haystack real

task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]

head-level needle mass: max 0.9973 p99 0.7947 median 0.003481 (257/1152 heads >= 0.05)

heads above 0.001:719 0.005:529 0.01:460 0.05:257 0.1:172 0.25:86 [ADVISORY - MIN_MASS uncalibrated, not enforced]

RETIRED median-ladder gate (not enforced): niah 3.656 b cont 3.799 b delta -0.153 b 35.0% of heads niah>cont

HEADS IN BAND 9.1% (gain over best corner >= 2.0x at 3 b/token)

ROUTED GAIN 1.22x (geometric mean of max(gain,1) over all heads)

in-band heads median 2.4x p90 3.0x

ladder width median 3.68 b IQR [3.23, 4.19] 100% of heads above 1.4

tau median 2.55 excl. sinks 2.53

top-1 weight median 0.3942 95% mass in 112 tokens (1.5% of ctx)

B=2b vs uniform 45.5x vs eviction 1.2x vs BEST corner 1.2x evicts 65%

B=3b vs uniform 26.7x vs eviction 1.0x vs BEST corner 1.0x evicts 50%

quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.715 [H2 answered]

heads where eviction beats a 1-bit tier: 100%

H0 — decision summary (7/7)

qwen3-8b (ctx 32,768, 1,152 heads)

INPUT VALIDITY PASS haystack real

task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]

head-level needle mass: max 0.9440 p99 0.7937 median 0.000721 (204/1152 heads >= 0.05)

heads above 0.001:535 0.005:426 0.01:368 0.05:204 0.1:120 0.25:66 [ADVISORY - MIN_MASS uncalibrated, not enforced]

RETIRED median-ladder gate (not enforced): niah 4.294 b cont 4.629 b delta -0.310 b 25.9% of heads niah>cont

HEADS IN BAND 6.1% (gain over best corner >= 2.0x at 3 b/token)

ROUTED GAIN 1.13x (geometric mean of max(gain,1) over all heads)

in-band heads median 2.3x p90 3.0x

ladder width median 4.40 b IQR [3.81, 5.56] 100% of heads above 1.4

tau median 3.03 excl. sinks 3.02

top-1 weight median 0.2160 95% mass in 284 tokens (0.9% of ctx)

B=2b vs uniform 53.1x vs eviction 1.1x vs BEST corner 1.1x evicts 67%

B=3b vs uniform 29.3x vs eviction 1.0x vs BEST corner 1.0x evicts 53%

quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.635 [H2 answered]

heads where eviction beats a 1-bit tier: 100%

heads where eviction beats a 2-bit tier: 57%

linearization check (predicted/measured gain @3b): 0.99 (1.0 = exact)

VERDICT: STOP – Only 6% of heads sit in the productive band and routing buys 1.13x overall. Uniform quantization

or eviction already captures the gain almost everywhere; SIEVE would be a reframing without a result.

qwen3-8b (ctx 40,960, 1,152 heads)

INPUT VALIDITY PASS haystack real

task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]

head-level needle mass: max 0.9498 p99 0.7034 median 0.000370 (176/1152 heads >= 0.05)

heads above 0.001:491 0.005:401 0.01:336 0.05:176 0.1:114 0.25:60 [ADVISORY - MIN_MASS uncalibrated, not enforced]

RETIRED median-ladder gate (not enforced): niah 4.488 b cont 5.061 b delta -0.432 b 22.5% of heads niah>cont

HEADS IN BAND 5.6% (gain over best corner >= 2.0x at 3 b/token)

ROUTED GAIN 1.13x (geometric mean of max(gain,1) over all heads)

in-band heads median 2.4x p90 2.9x

ladder width median 4.65 b IQR [3.96, 6.15] 100% of heads above 1.4

tau median 3.21 excl. sinks 3.20

top-1 weight median 0.1813 95% mass in 413 tokens (1.1% of ctx)

B=2b vs uniform 53.3x vs eviction 1.0x vs BEST corner 1.0x evicts 67%

B=3b vs uniform 29.9x vs eviction 1.0x vs BEST corner 1.0x evicts 53%

quantizer alpha @2b 0.940 (1.0 = no temperature distortion); spearman@2b 0.628 [H2 answered]

heads where eviction beats a 1-bit tier: 100%

heads where eviction beats a 2-bit tier: 60%

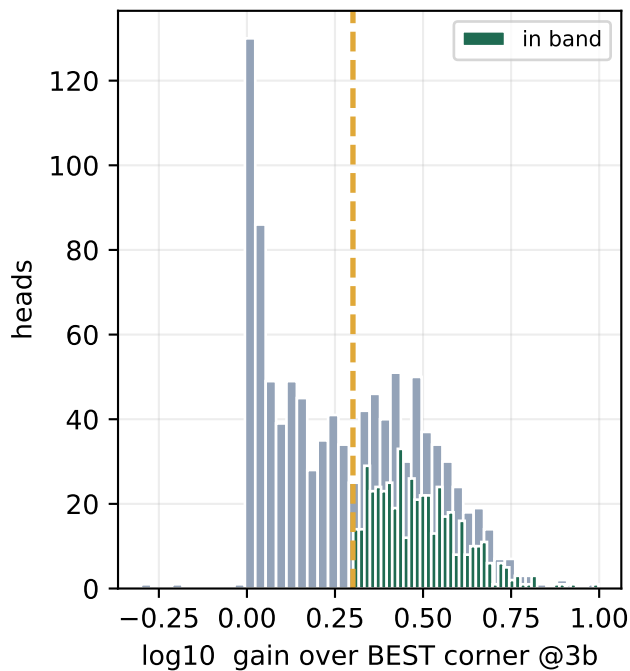
linearization check (predicted/measured gain @3b): 0.98 (1.0 = exact)

VERDICT: STOP – Only 6% of heads sit in the productive band and routing buys 1.13x overall. Uniform quantization

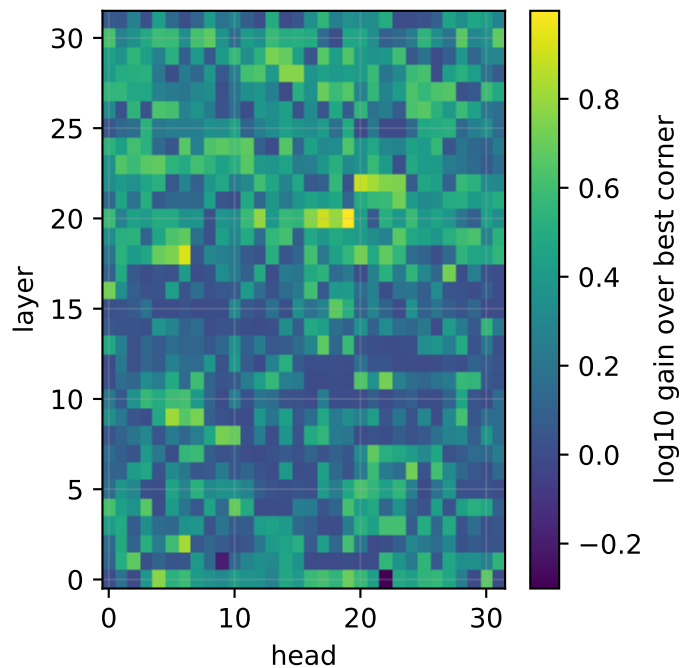
or eviction already captures the gain almost everywhere; SIEVE would be a reframing without a result.

llama31-8b — ctx 4,096

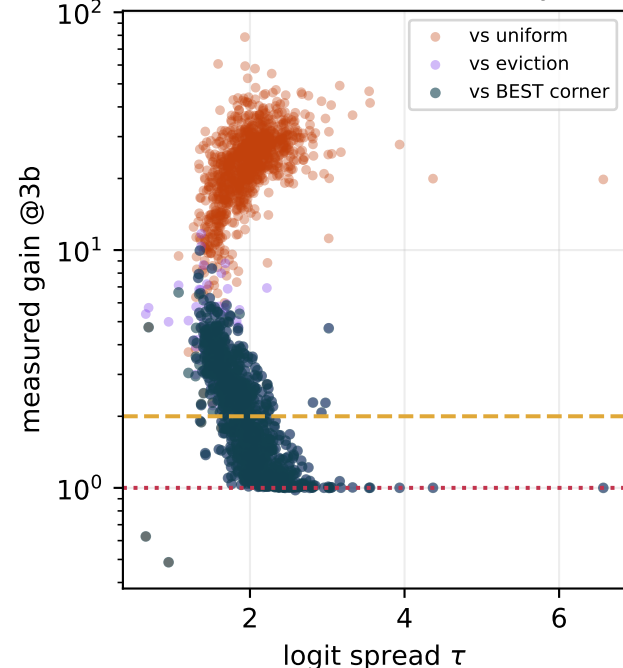
HEADS IN BAND: 46%



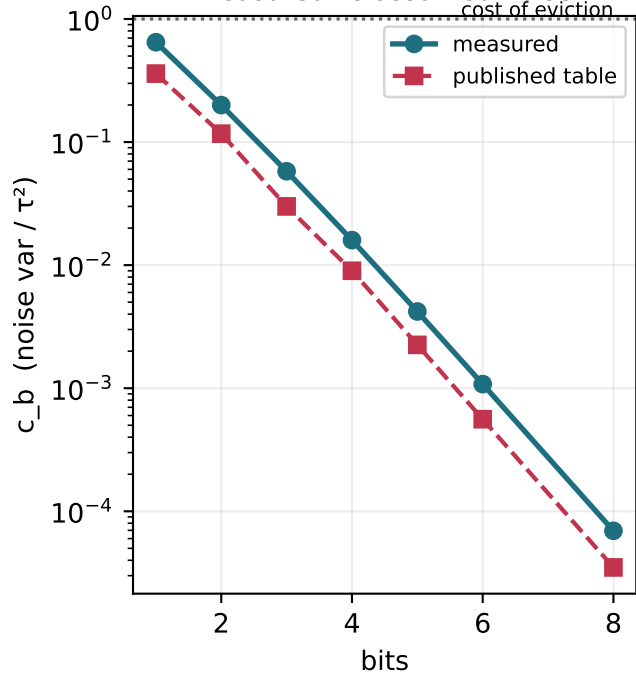
Which heads the router sends to SIEVE



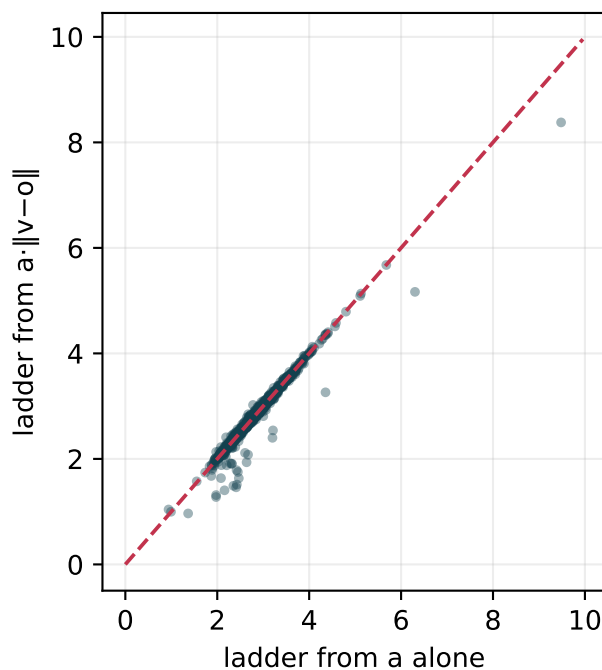
Where does the interior actually win?



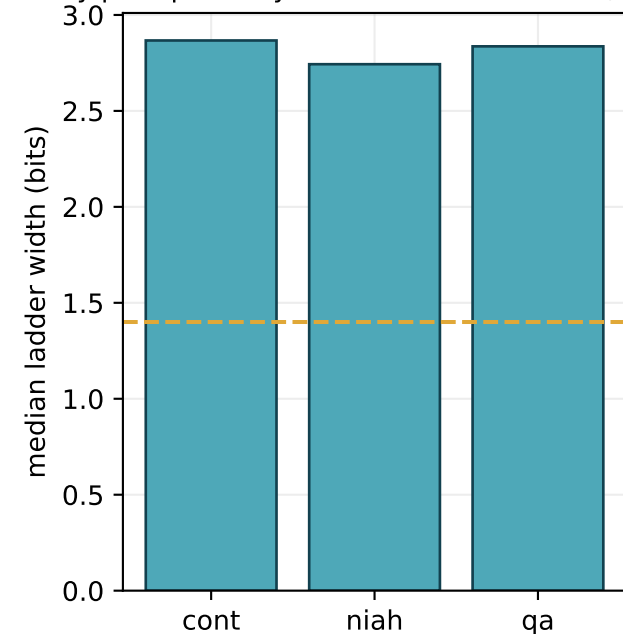
Measured vs assumed noise



Does the value term matter?

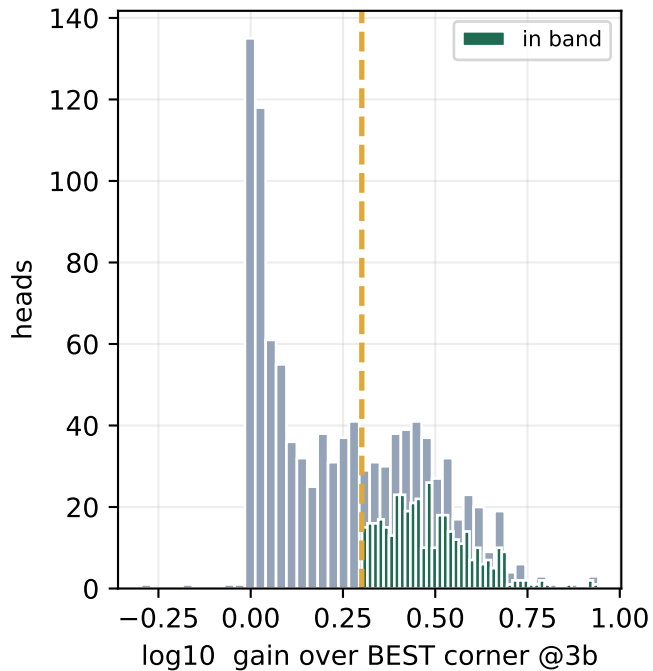


By prompt family — niah—cont -0.148 b, PA

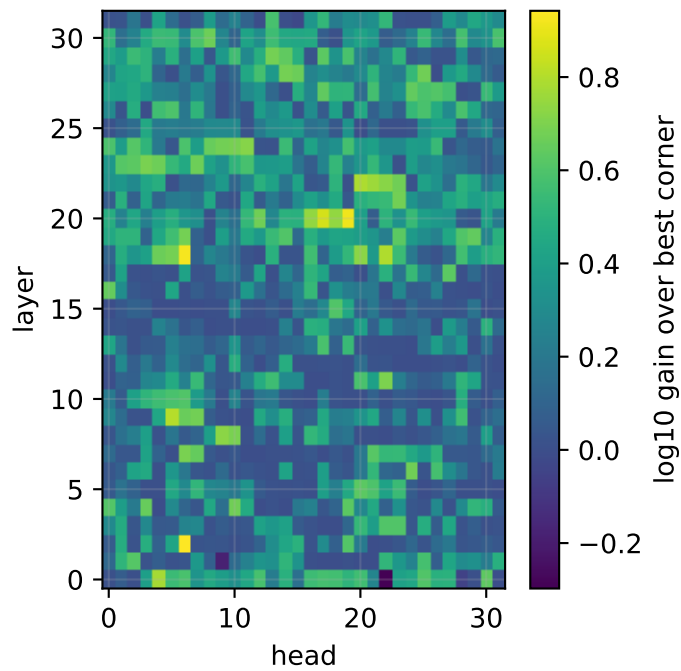


llama31-8b — ctx 8,192

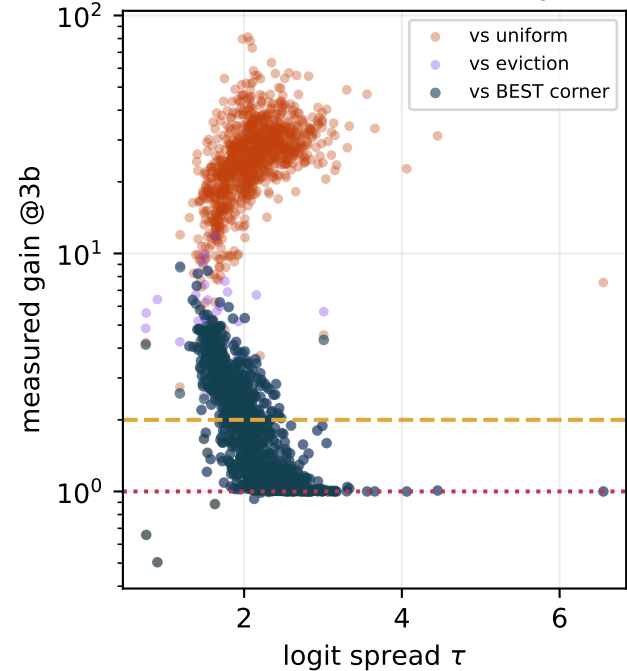
HEADS IN BAND: 39%



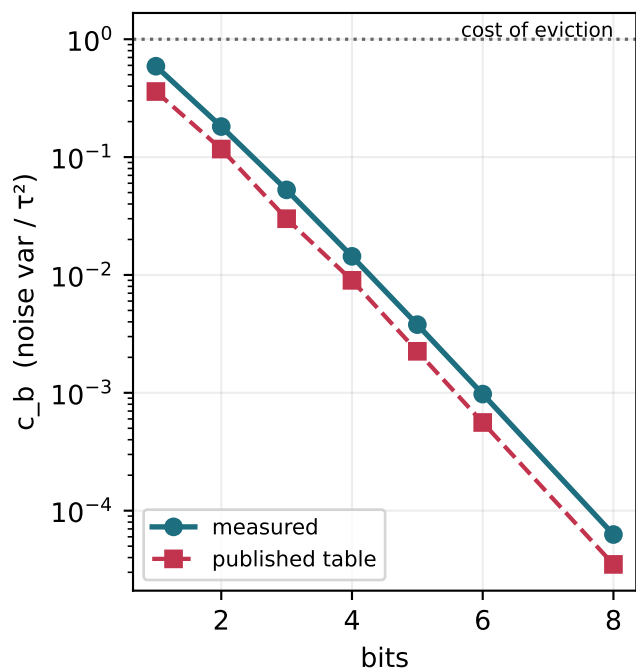
Which heads the router sends to SIEVE



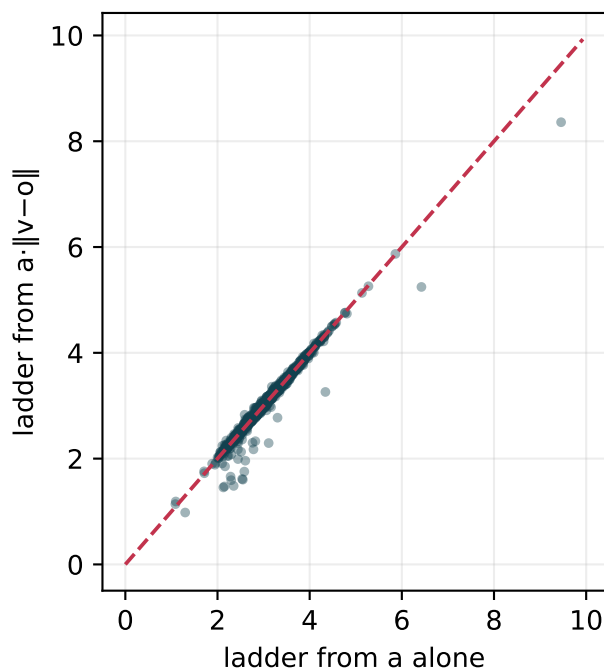
Where does the interior actually win?



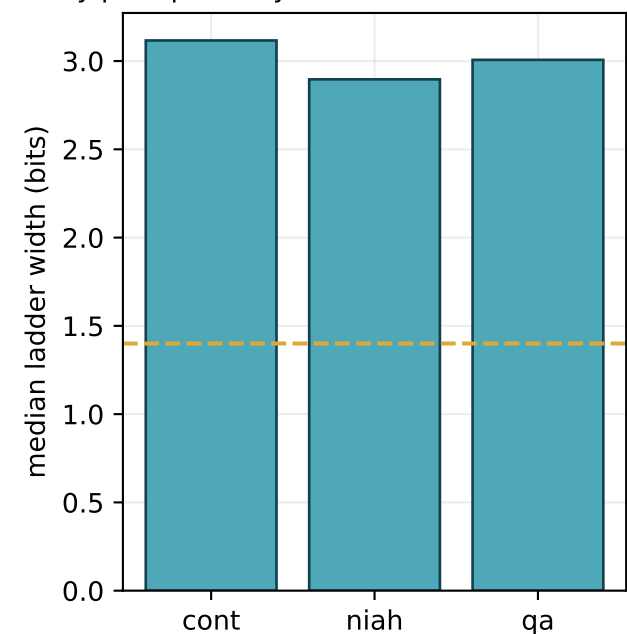
Measured vs assumed noise



Does the value term matter?

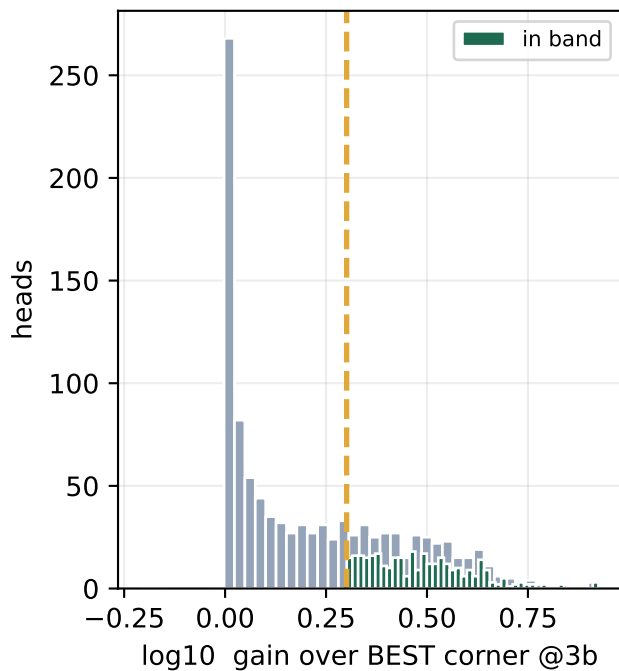


By prompt family — niah—cont -0.231 b, PA

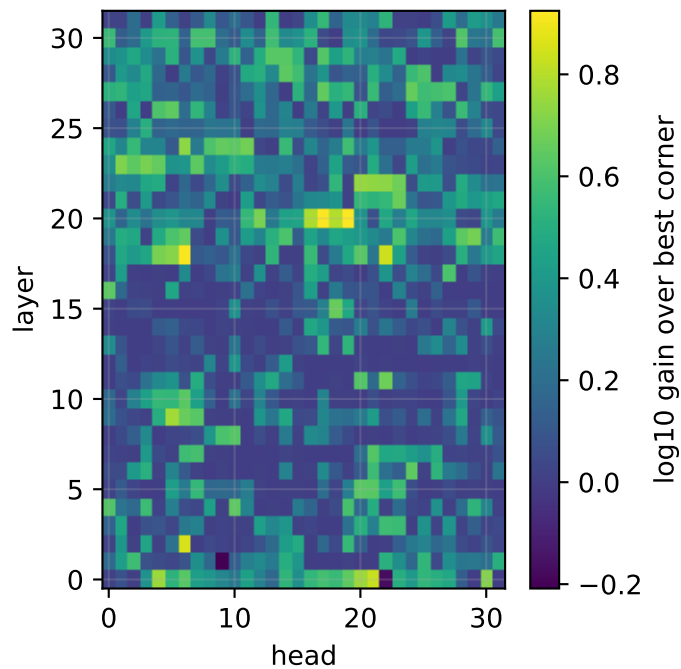


llama31-8b — ctx 16,384

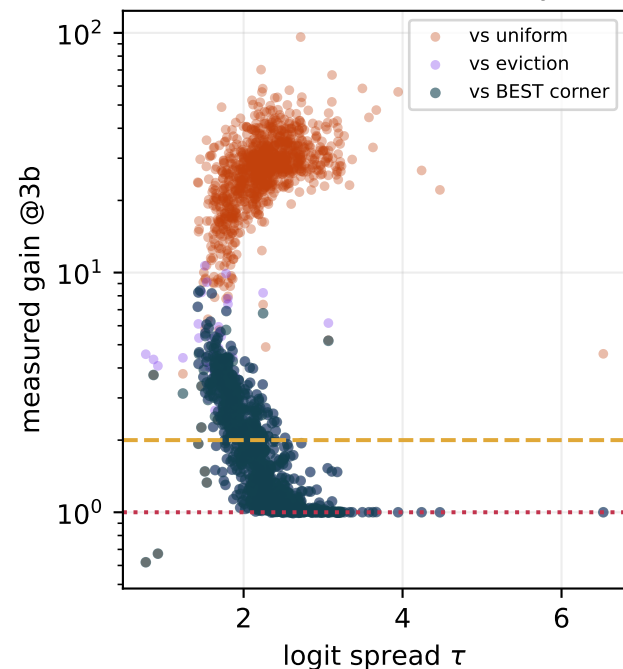
HEADS IN BAND: 33%



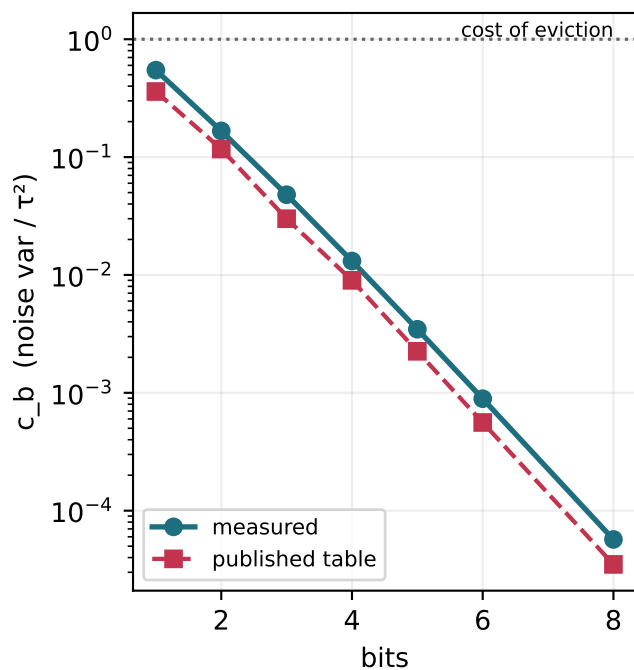
Which heads the router sends to SIEVE



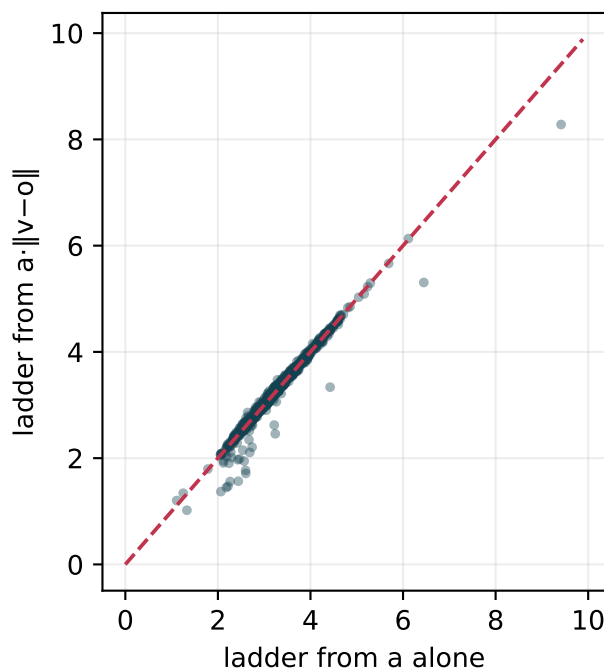
Where does the interior actually win?



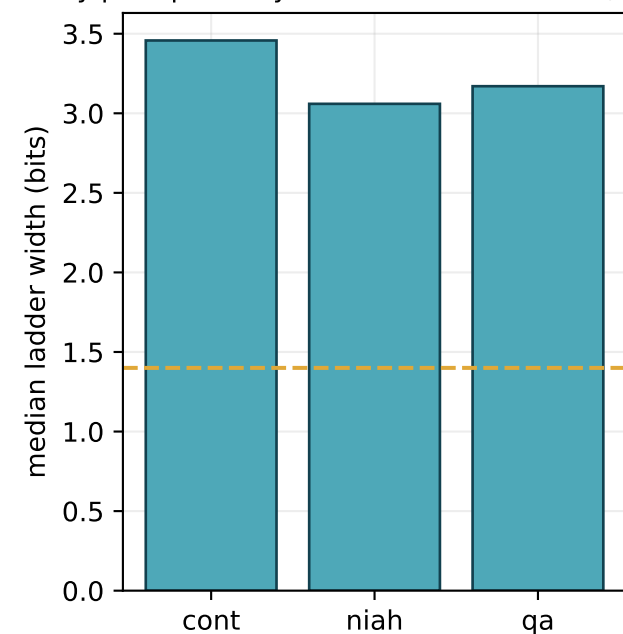
Measured vs assumed noise



Does the value term matter?

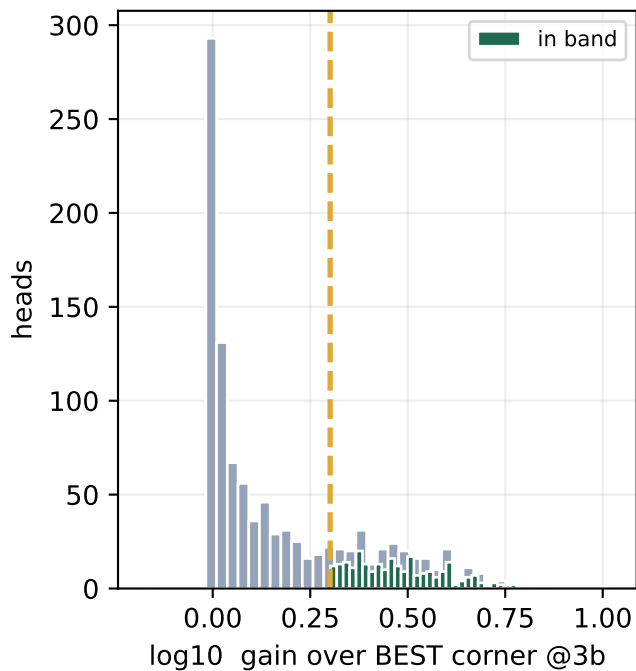


By prompt family — niah-cont -0.345 b, PA

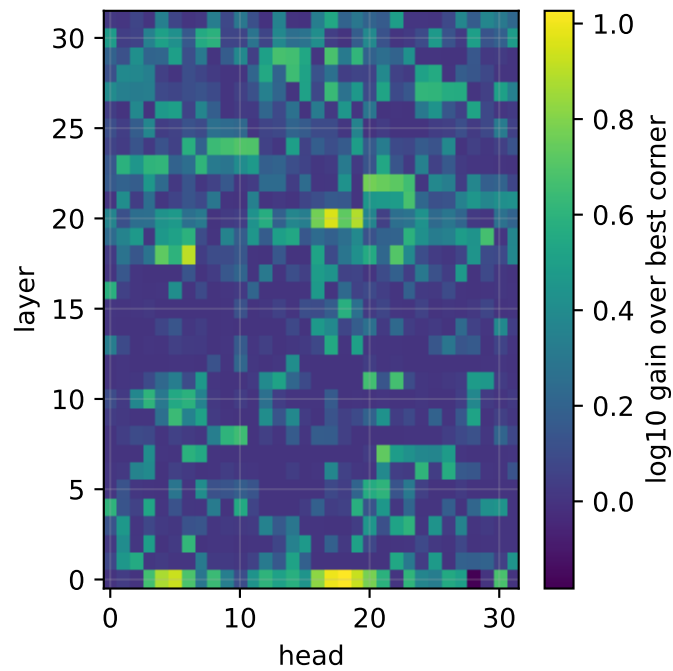


llama31-8b — ctx 32,768

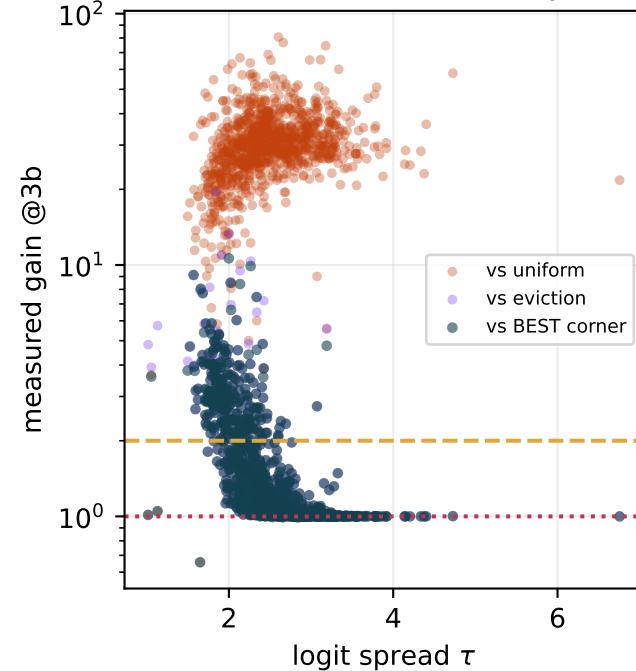
HEADS IN BAND: 26%



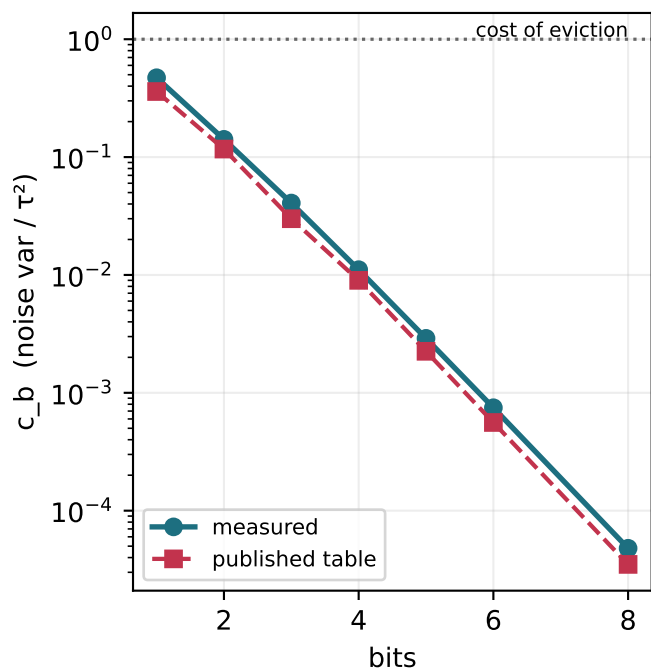
Which heads the router sends to SIEVE



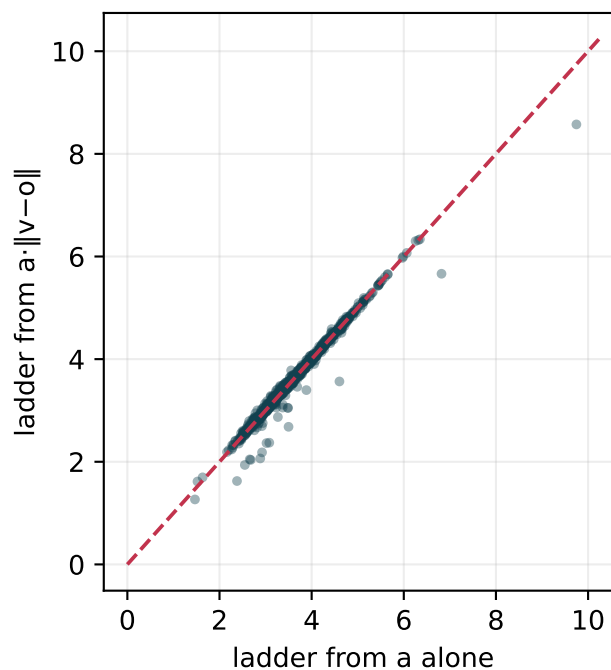
Where does the interior actually win?



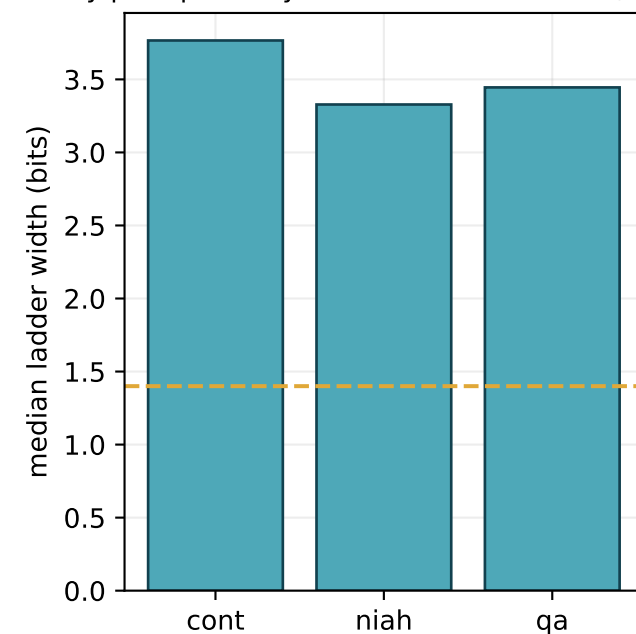
Measured vs assumed noise



Does the value term matter?

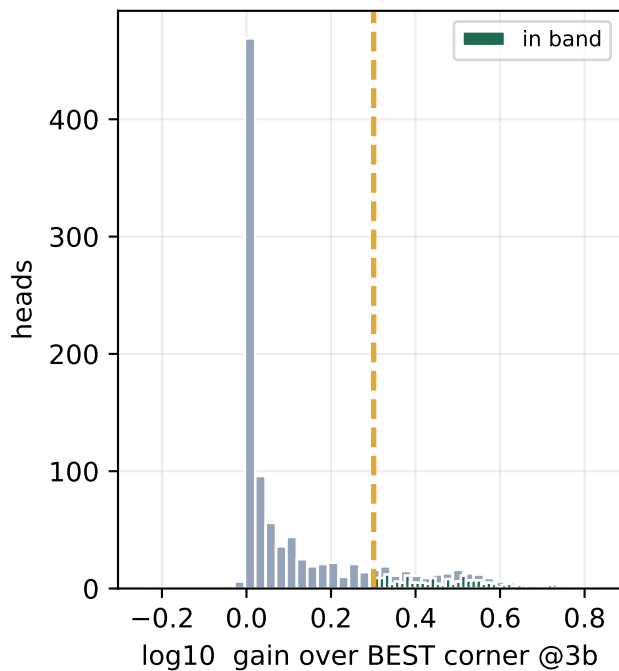


By prompt family — niah-cont -0.418 b, PA

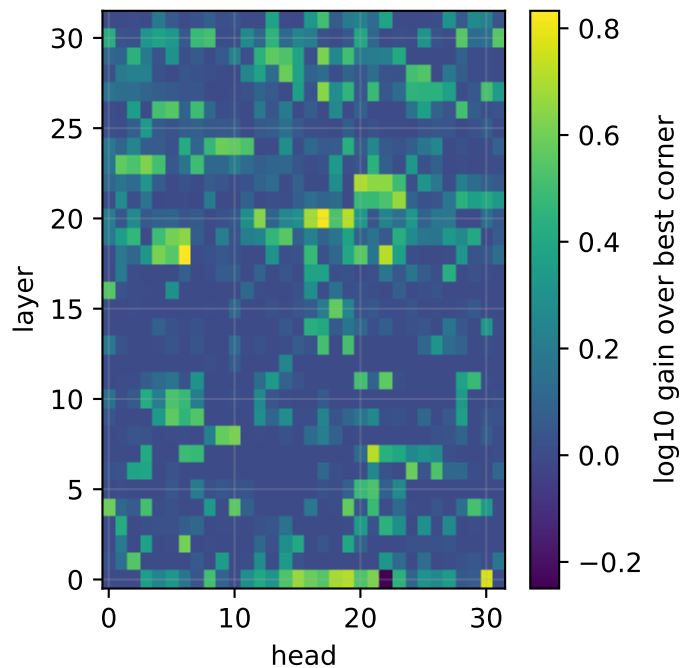


llama31-8b — ctx 65,536

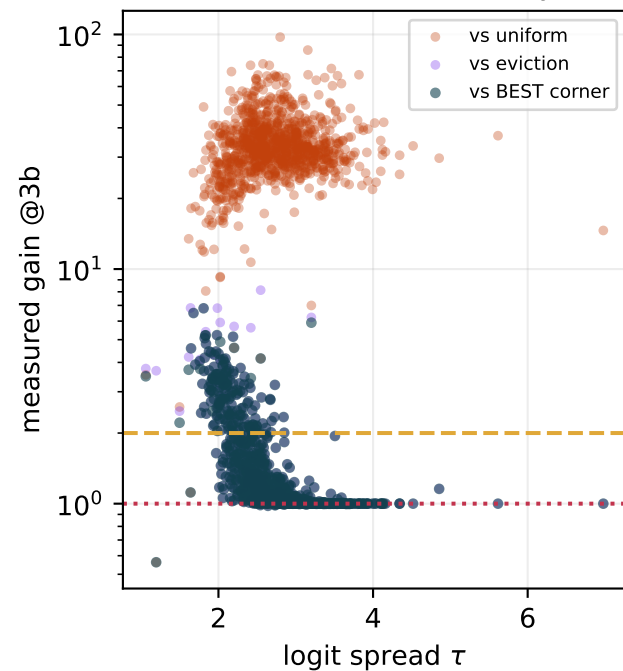
HEADS IN BAND: 17%



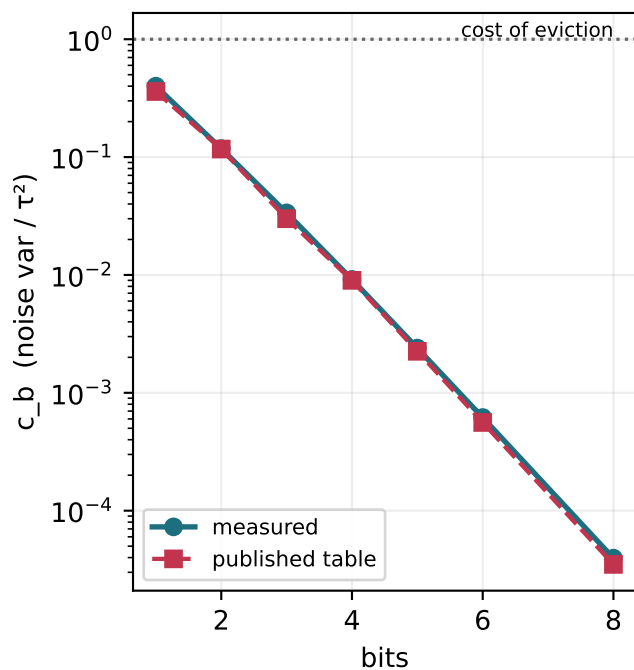
Which heads the router sends to SIEVE



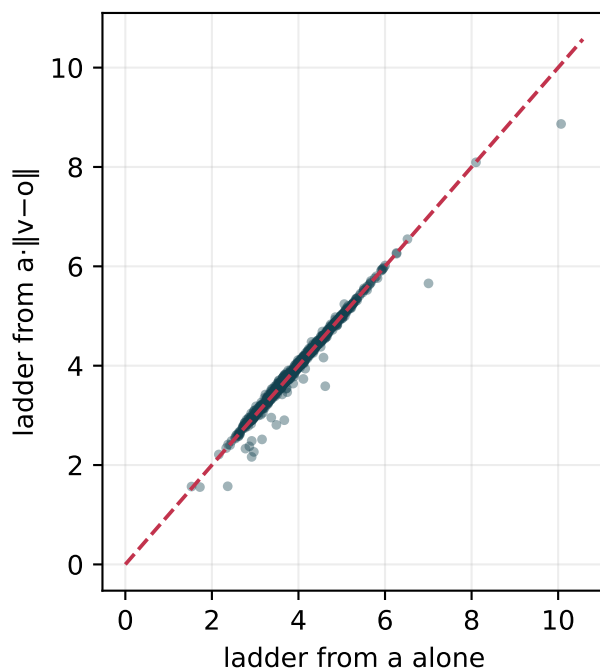
Where does the interior actually win?



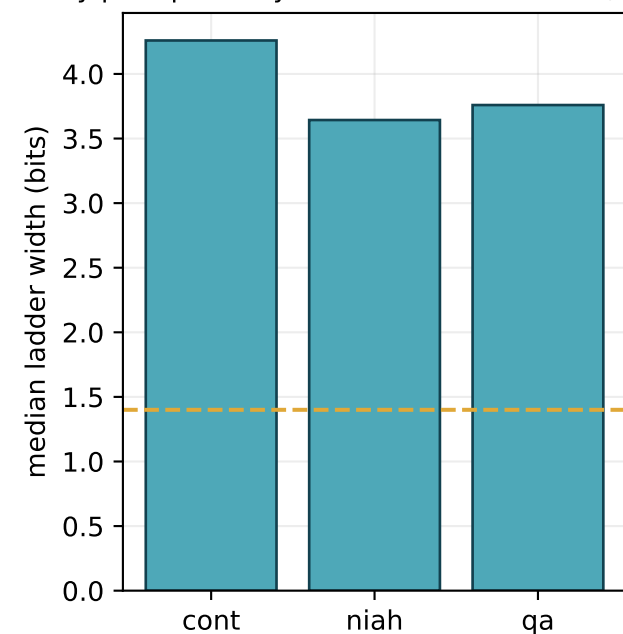
Measured vs assumed noise



Does the value term matter?

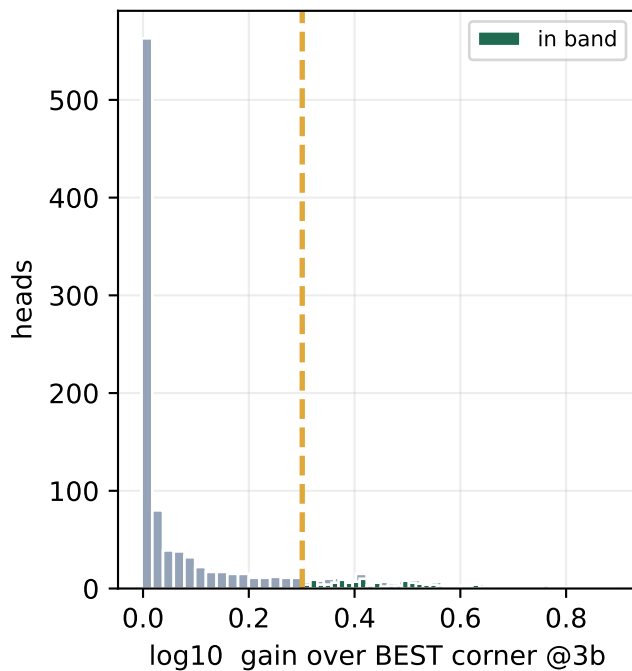


By prompt family — niah—cont -0.554 b, PA

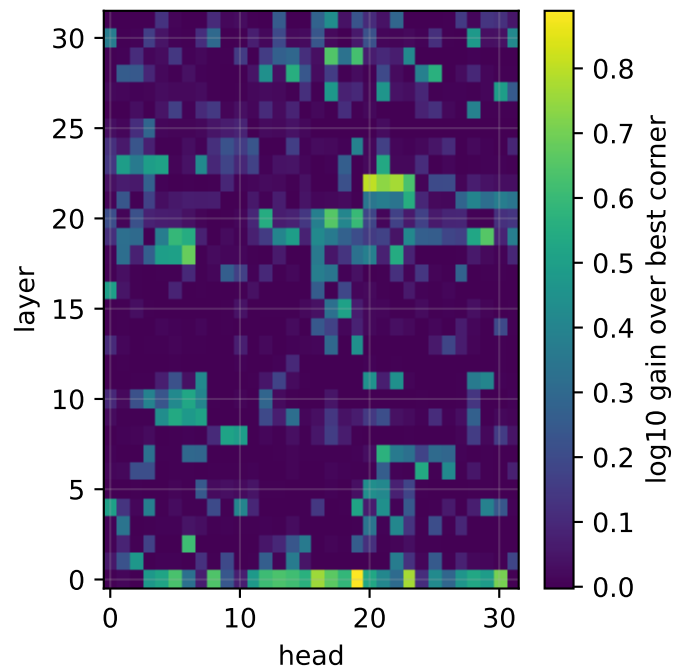


llama31-8b — ctx 131,072

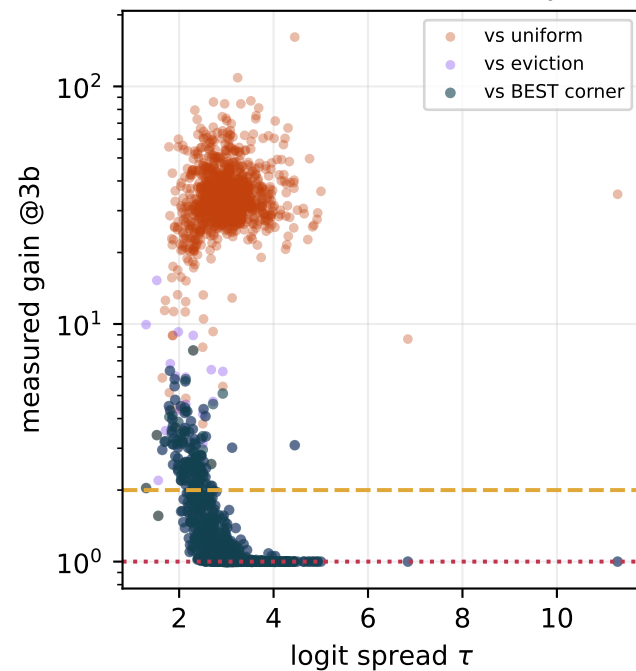
HEADS IN BAND: 13%



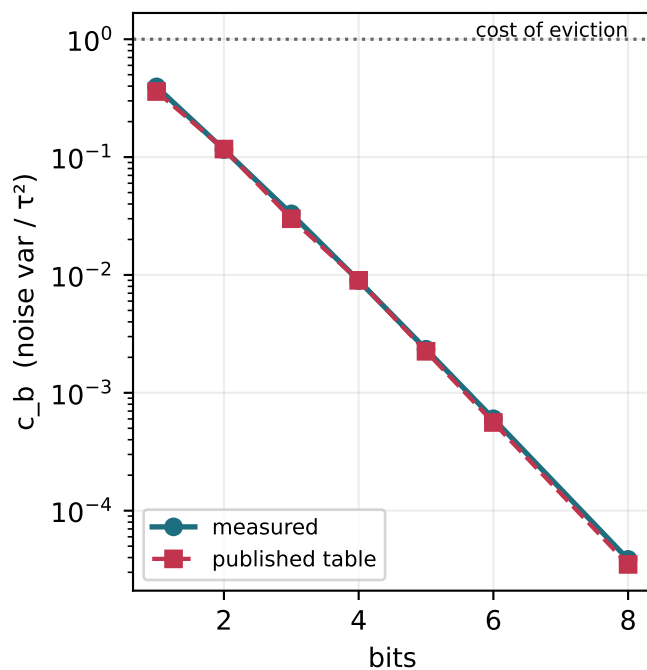
Which heads the router sends to SIEVE



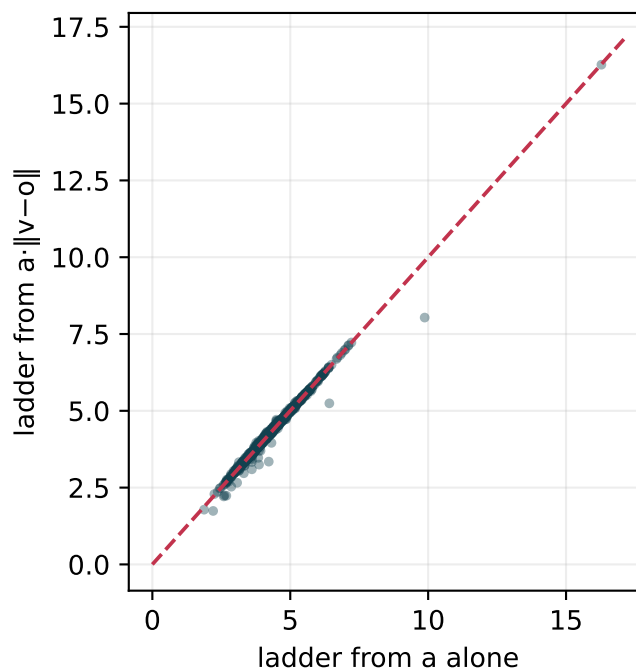
Where does the interior actually win?



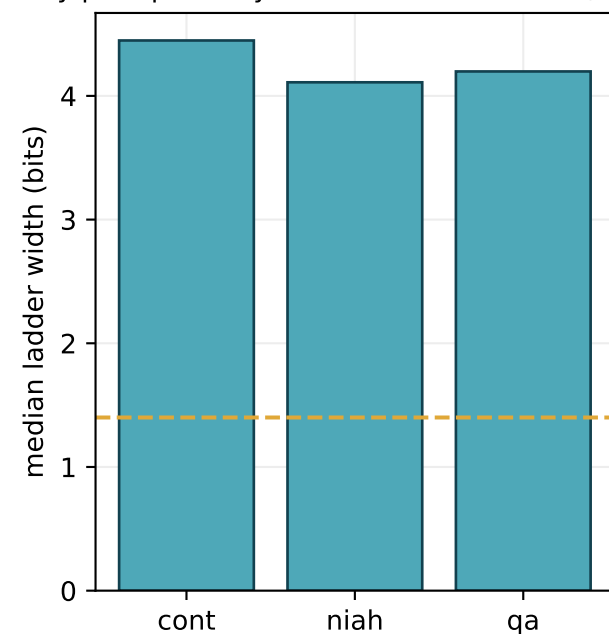
Measured vs assumed noise



Does the value term matter?

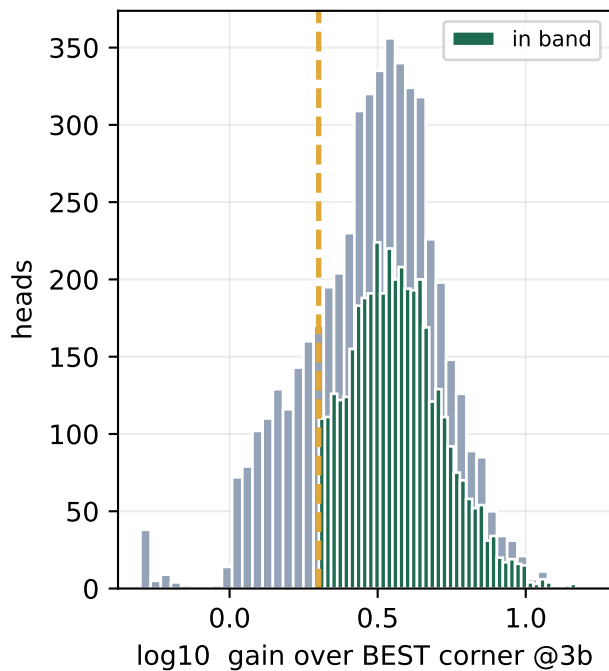


By prompt family — niah—cont -0.352 b, PA

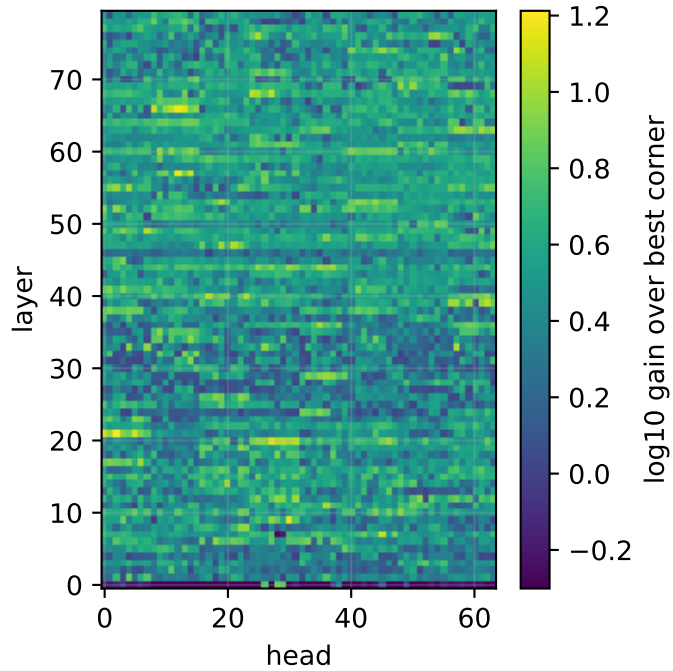


llama33-70b — ctx 8,192

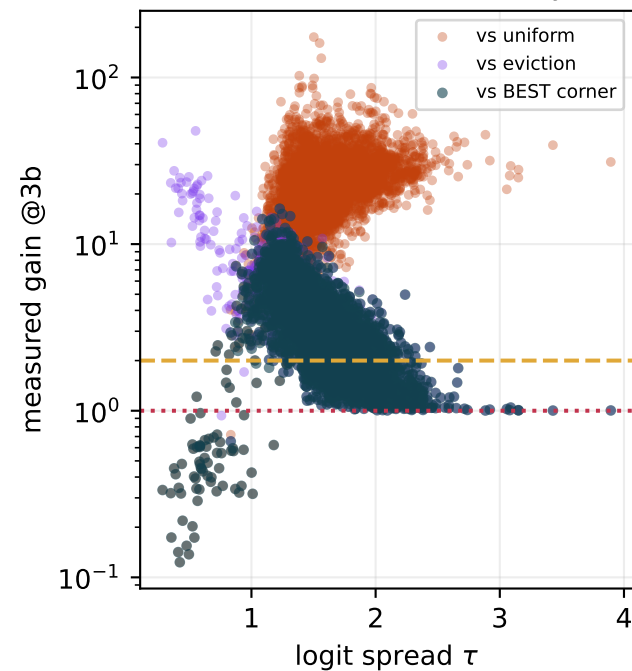
HEADS IN BAND: 79%



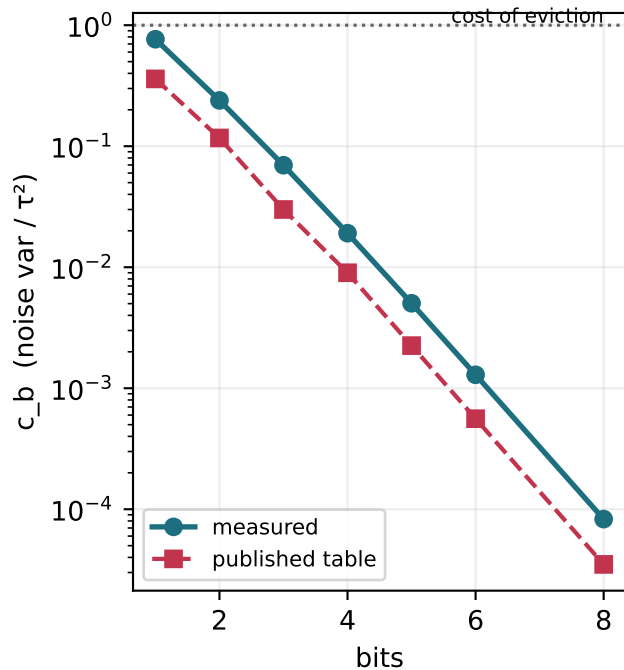
Which heads the router sends to SIEVE



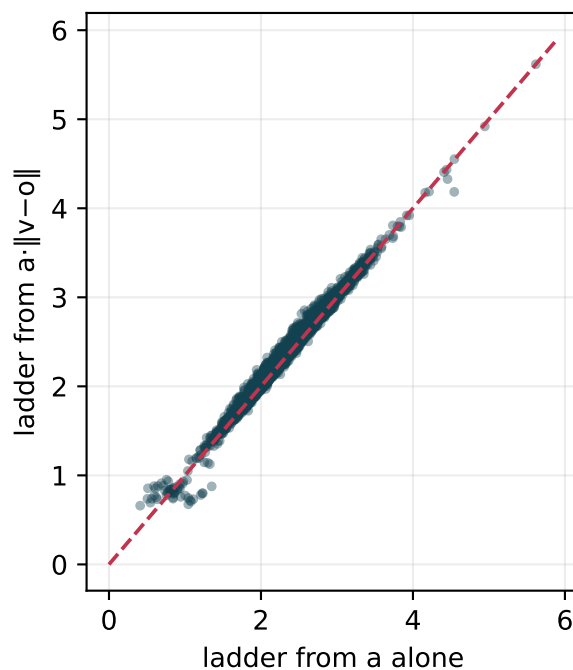
Where does the interior actually win?



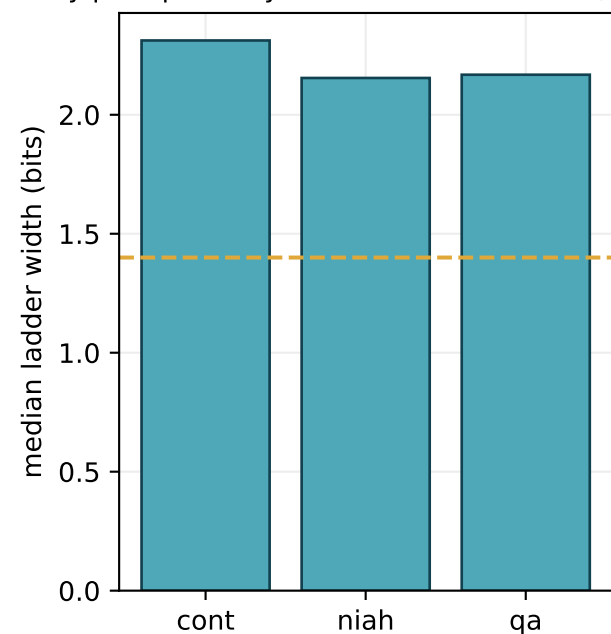
Measured vs assumed noise



Does the value term matter?

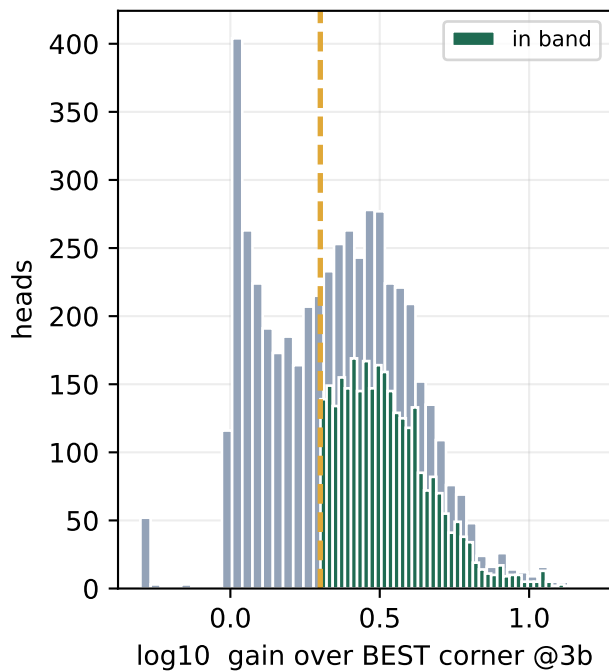


By prompt family — niah—cont -0.149 b, PA

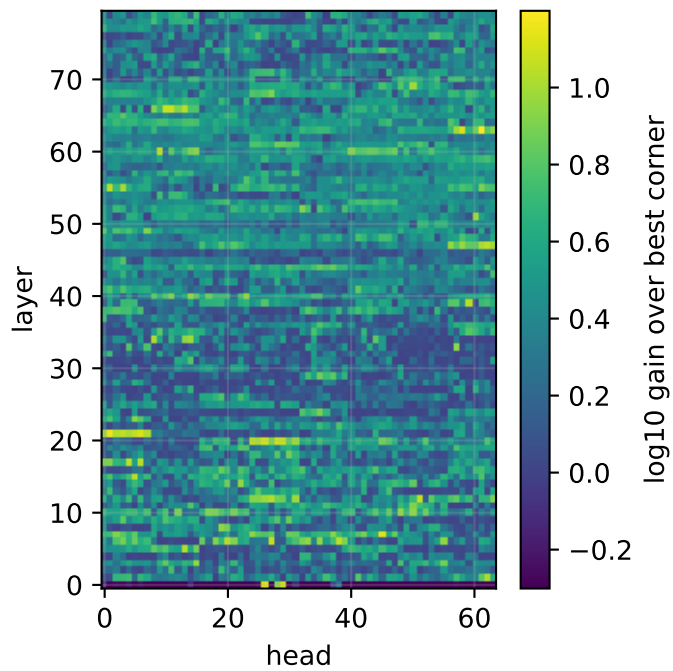


llama33-70b — ctx 32,768

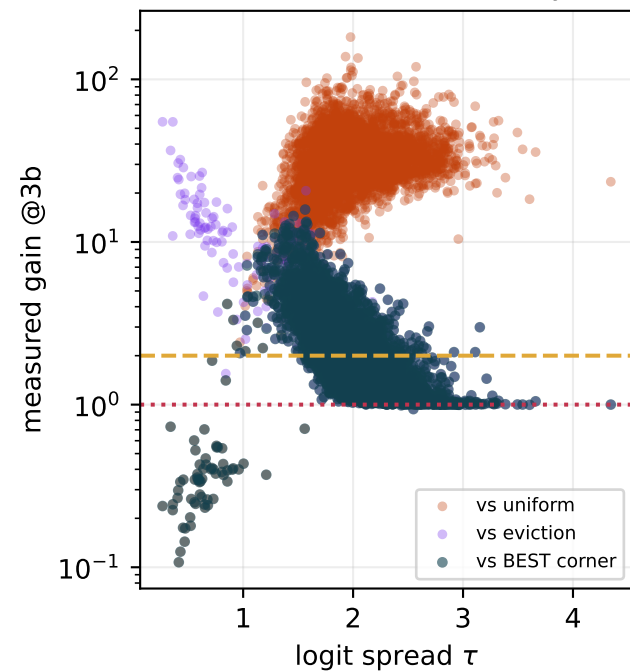
HEADS IN BAND: 58%



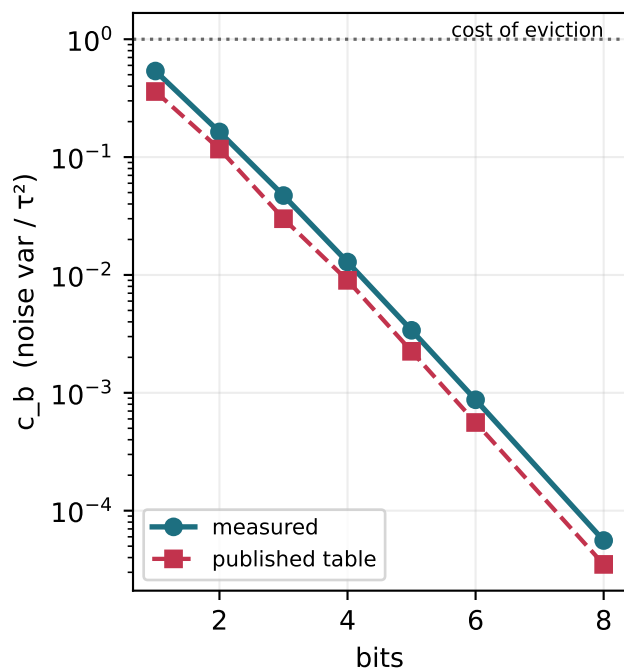
Which heads the router sends to SIEVE



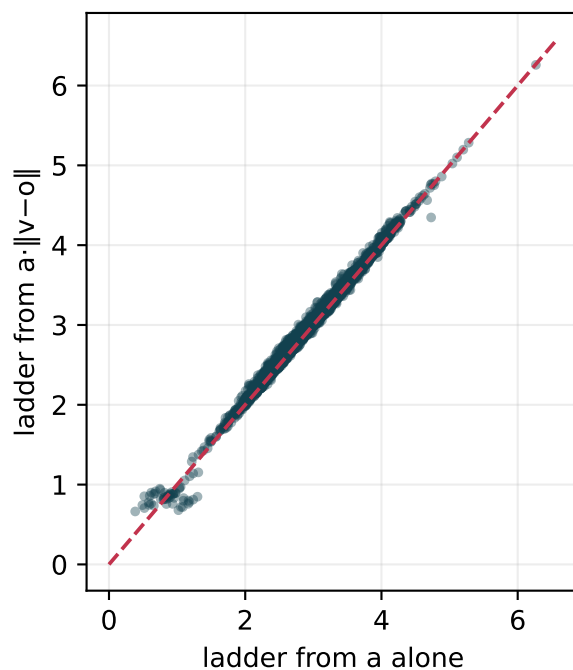
Where does the interior actually win?



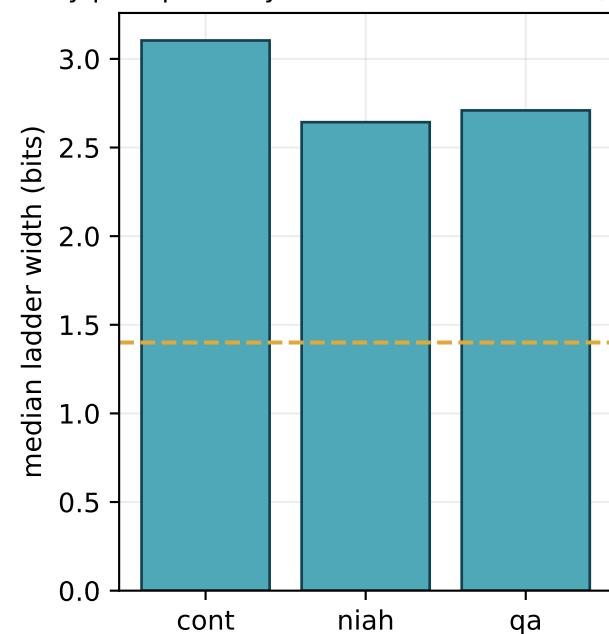
Measured vs assumed noise



Does the value term matter?

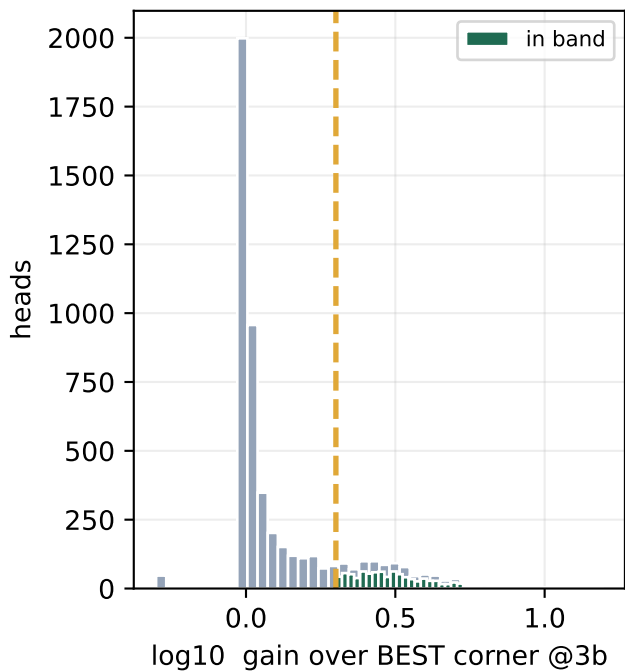


By prompt family — niah—cont -0.425 b, PA

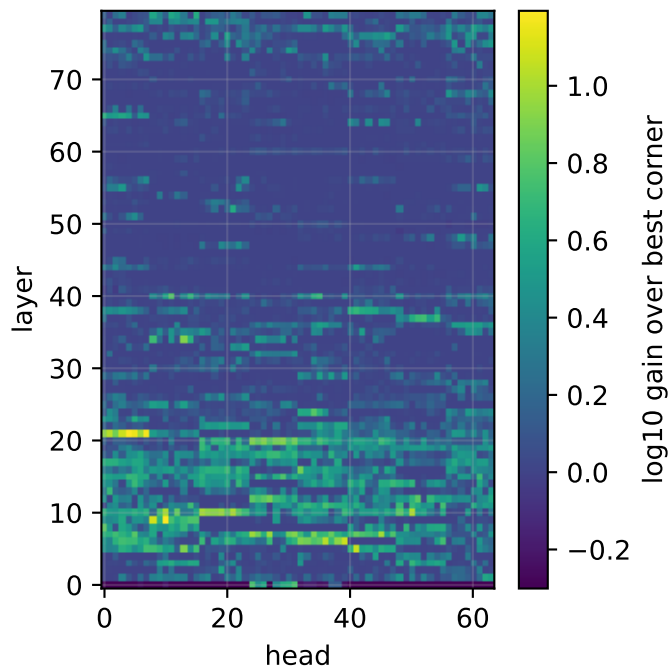


llama33-70b — ctx 131,072

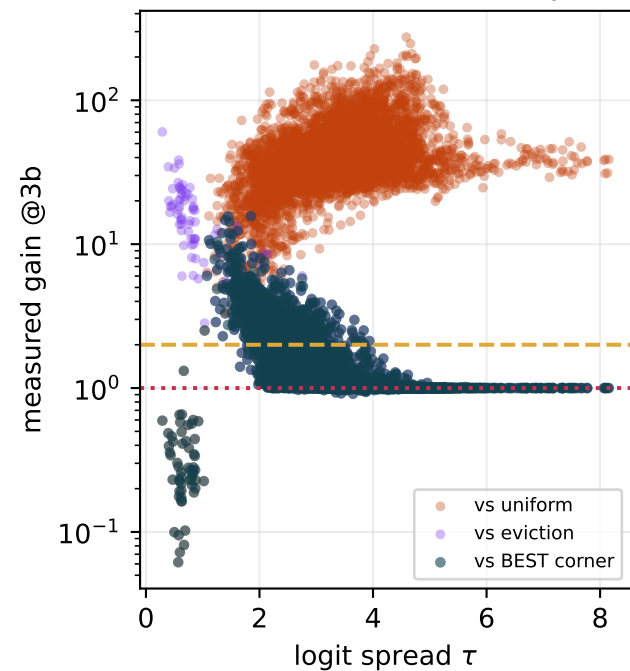
HEADS IN BAND: 18%



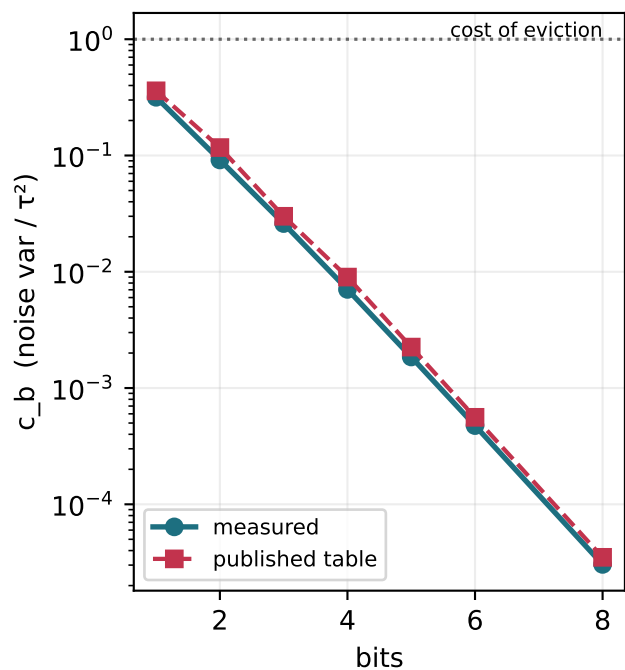
Which heads the router sends to SIEVE



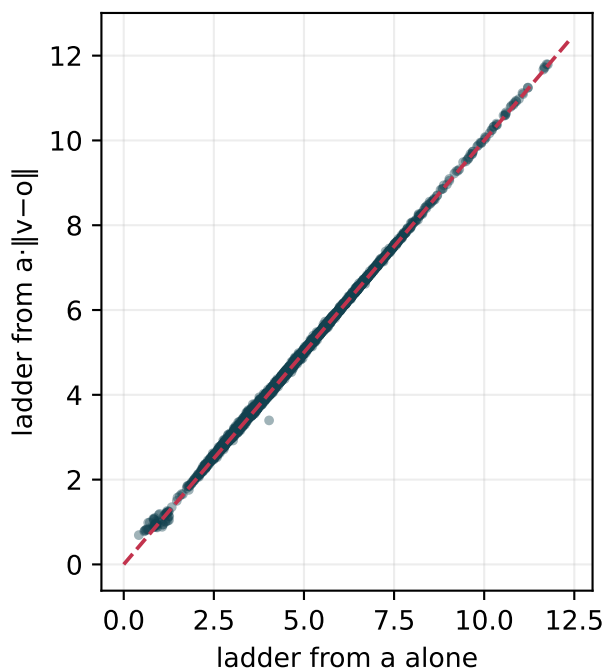
Where does the interior actually win?



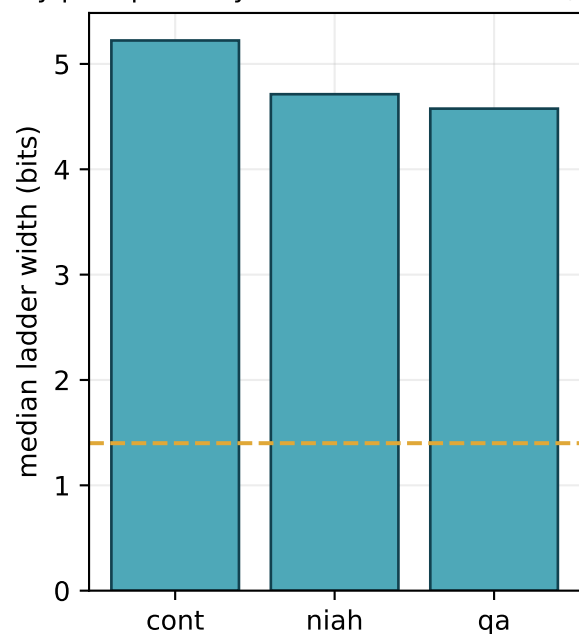
Measured vs assumed noise



Does the value term matter?

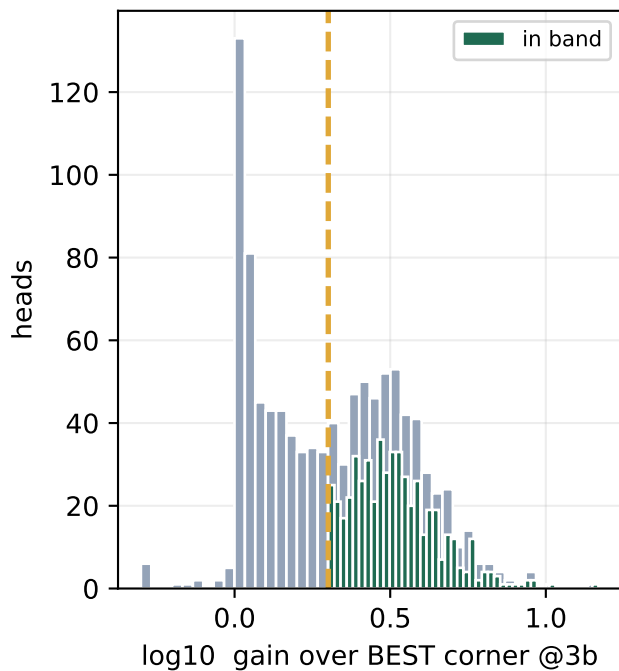


By prompt family — niah-cont -0.379 b, PA

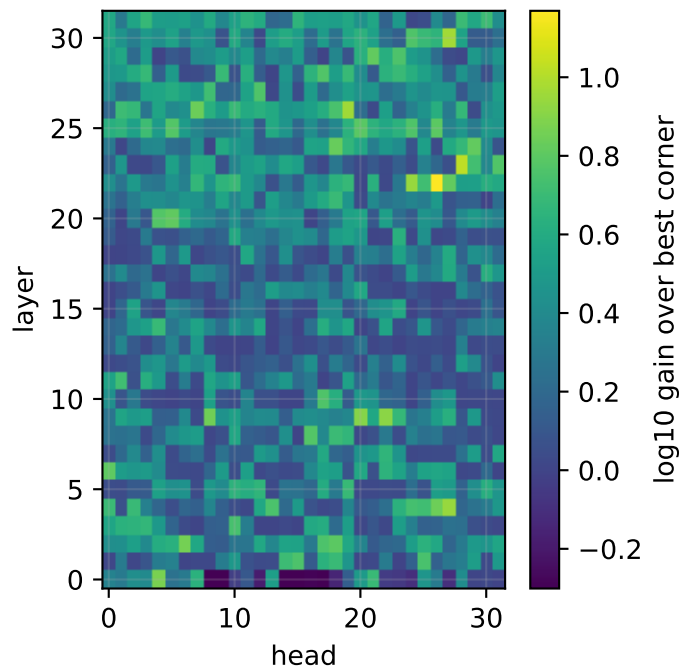


mistral-7b — ctx 8,192

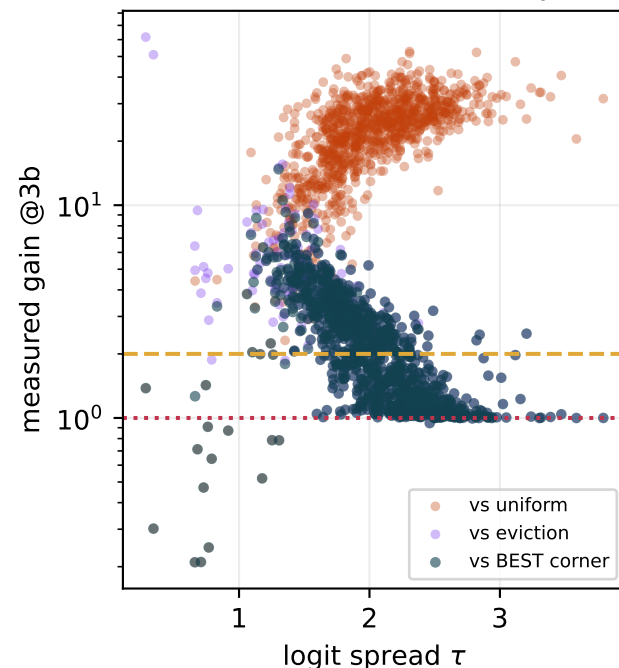
HEADS IN BAND: 51%



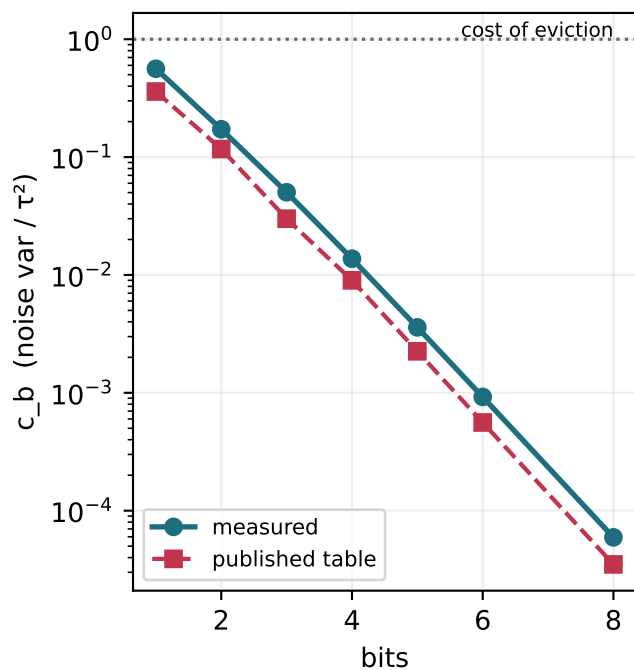
Which heads the router sends to SIEVE



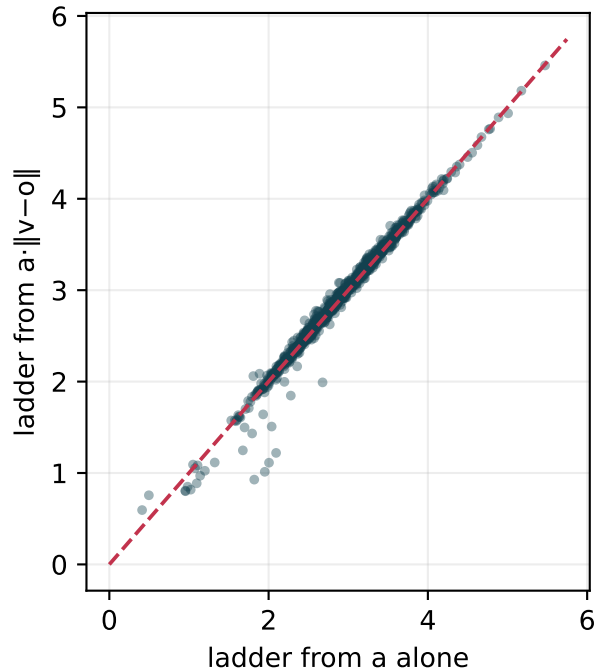
Where does the interior actually win?



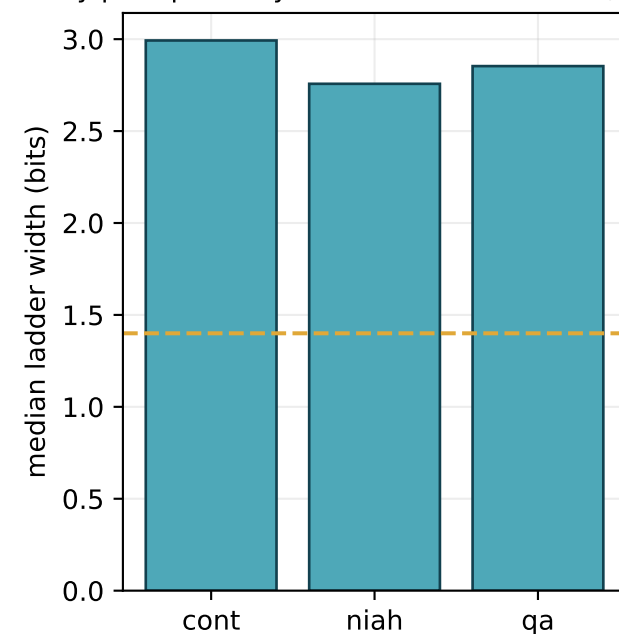
Measured vs assumed noise



Does the value term matter?

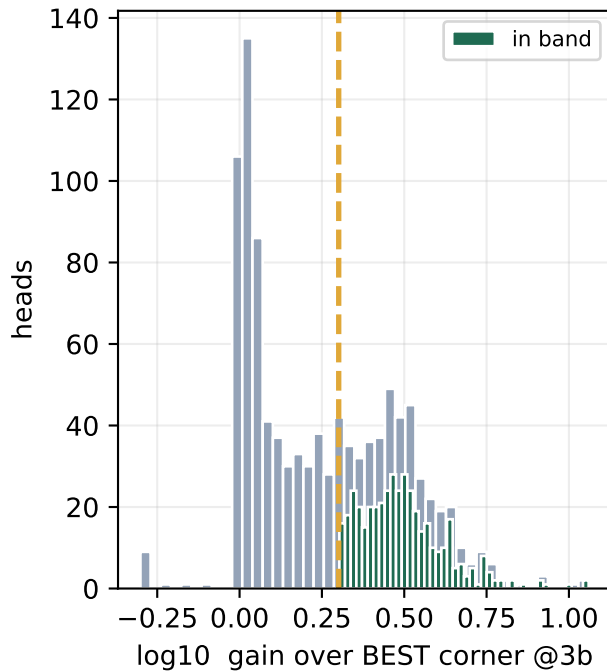


By prompt family — niah-cont -0.200 b, PA

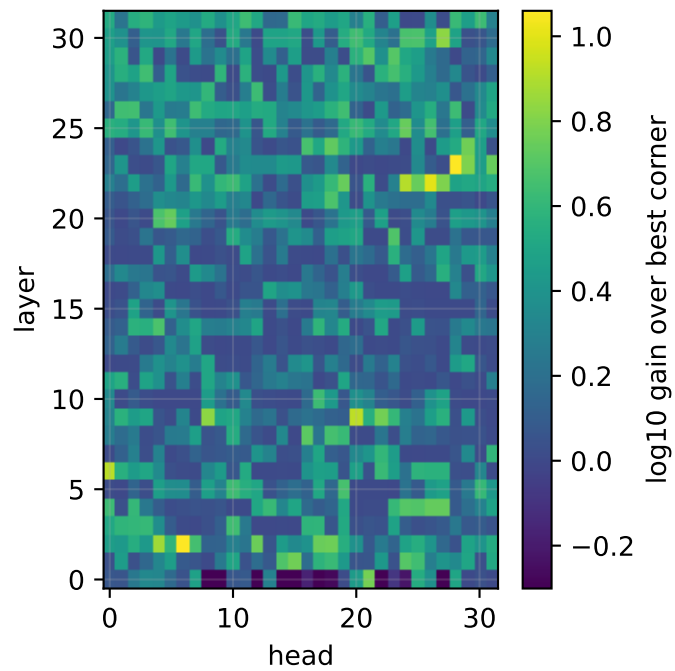


mistral-7b — ctx 32,768

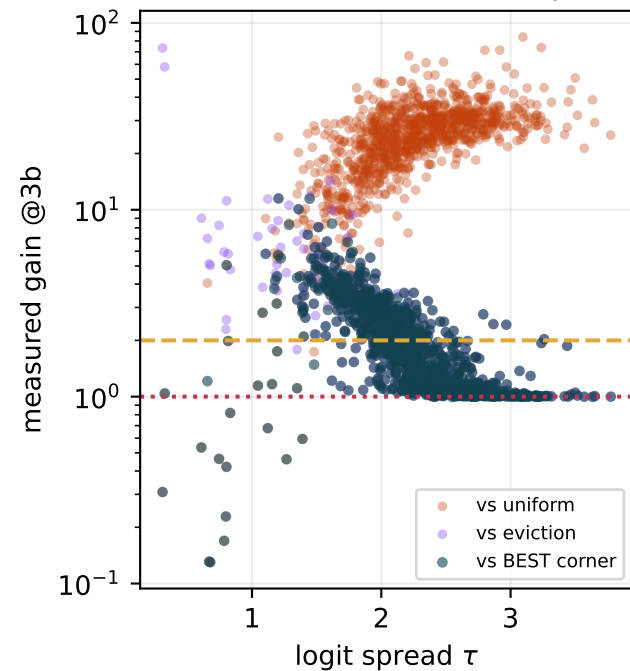
HEADS IN BAND: 41%



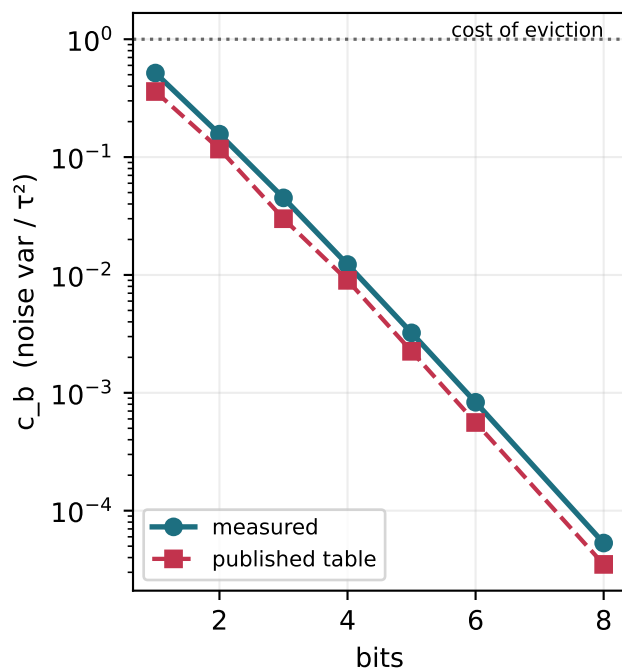
Which heads the router sends to SIEVE



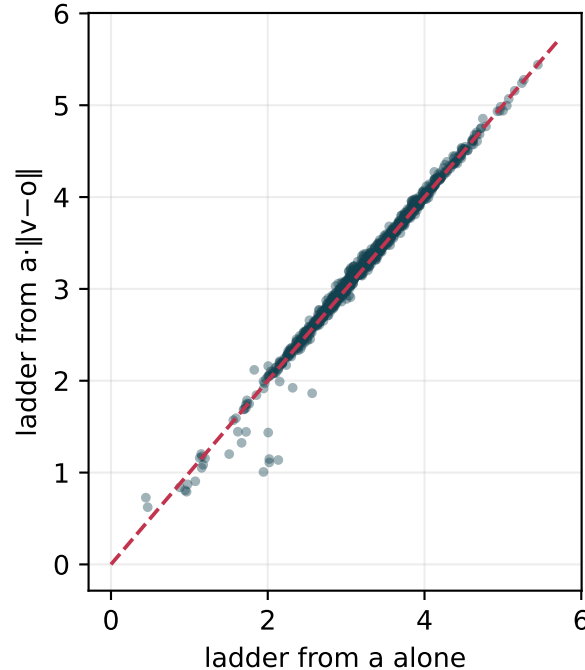
Where does the interior actually win?



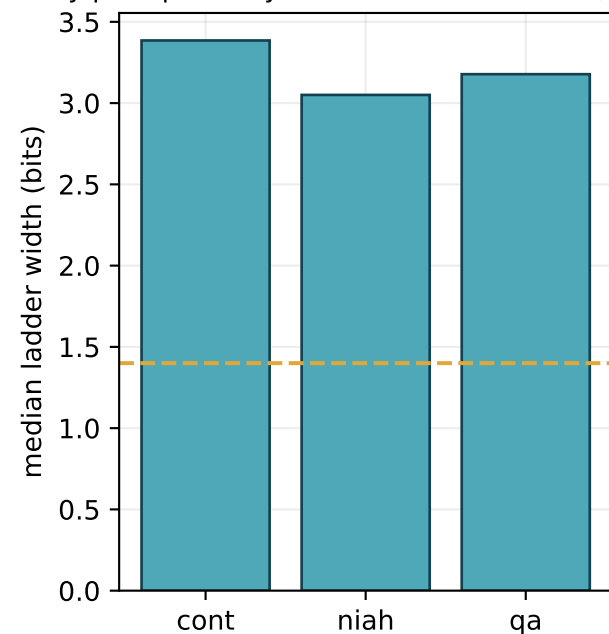
Measured vs assumed noise



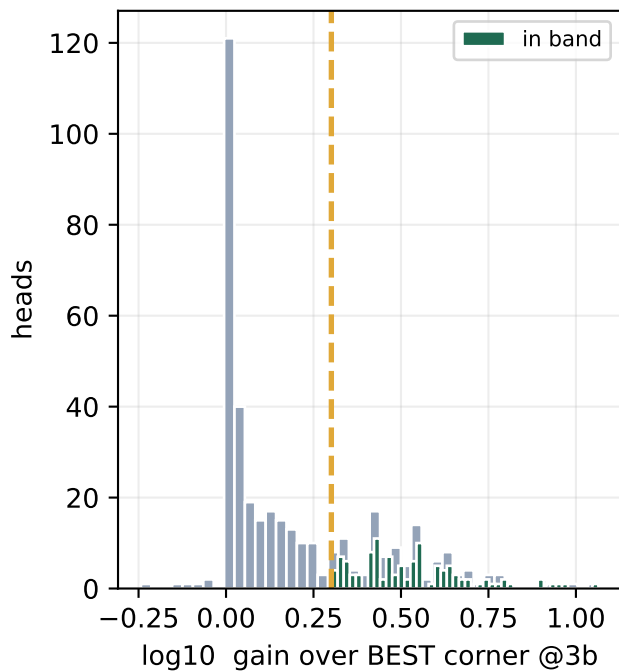
Does the value term matter?



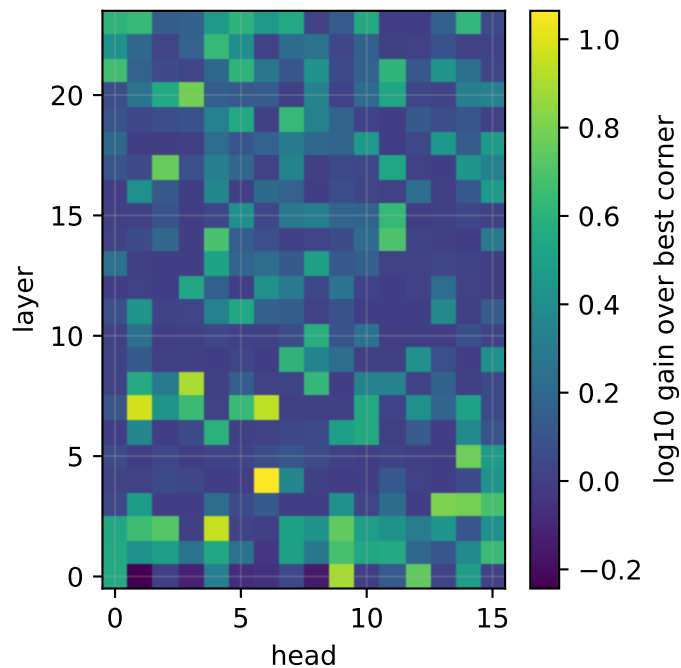
By prompt family — niah-cont -0.296 b, PA



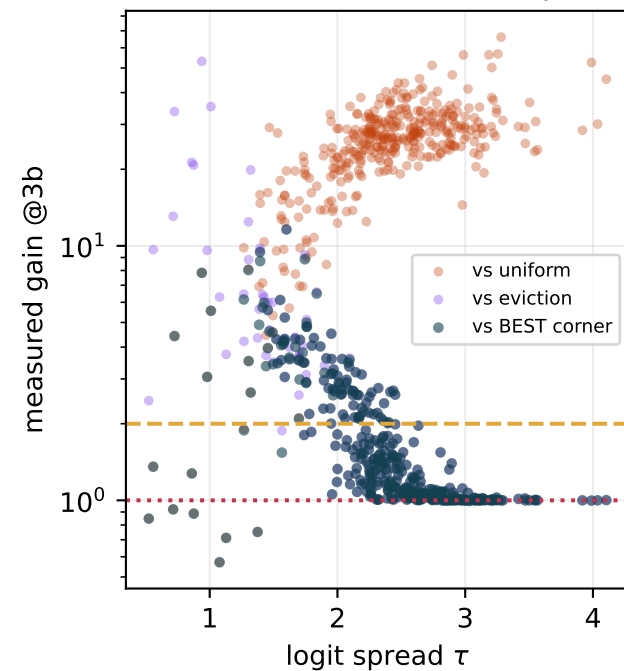
HEADS IN BAND: 29%



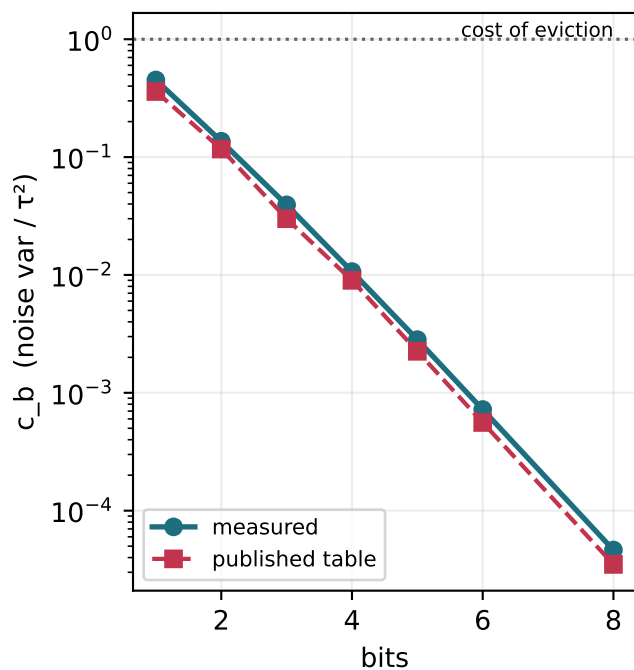
Which heads the router sends to SIEVE



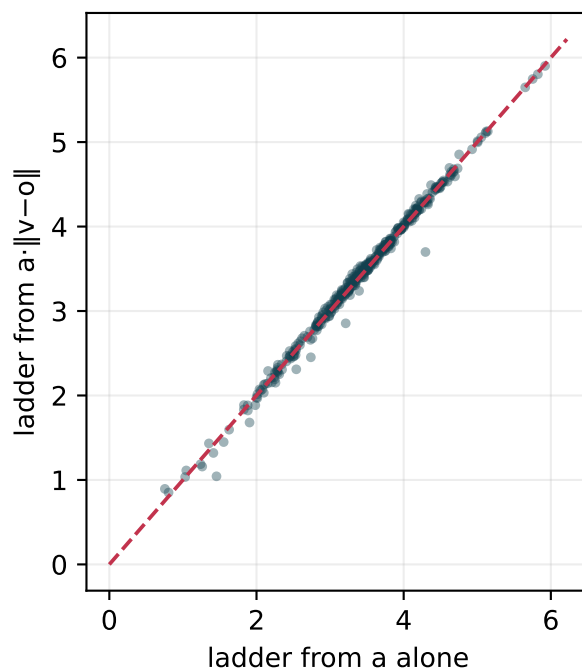
Where does the interior actually win?



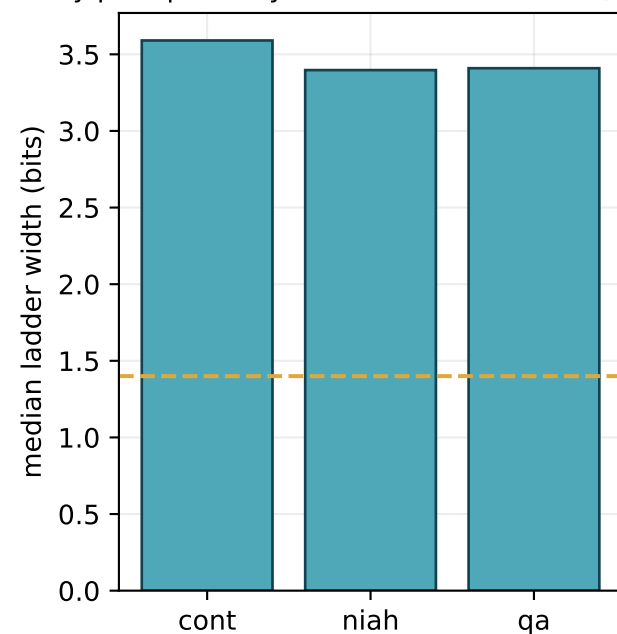
Measured vs assumed noise



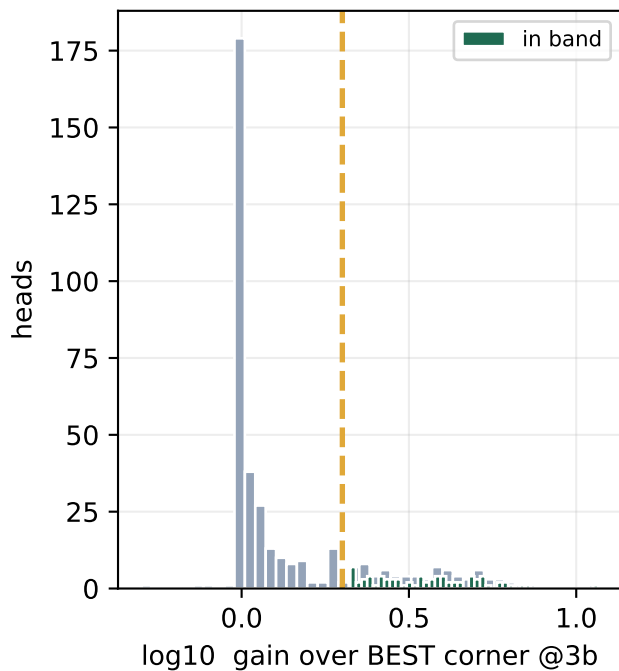
Does the value term matter?



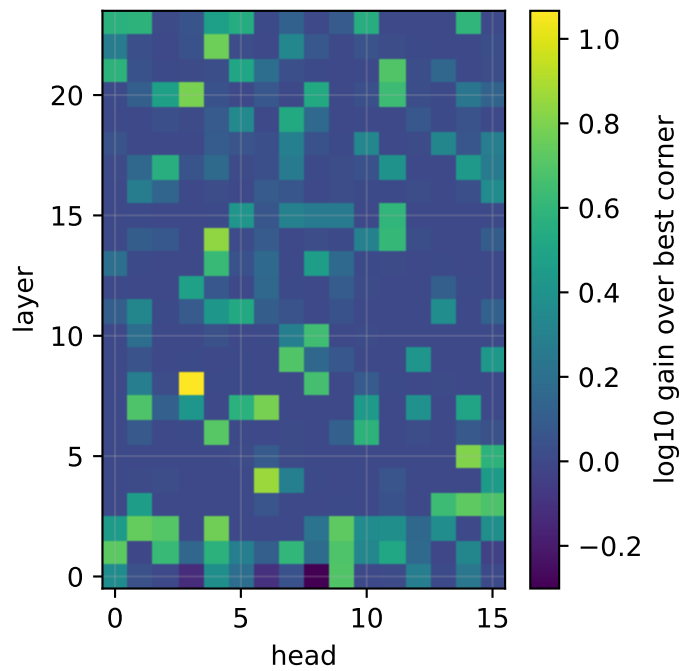
By prompt family — niah-cont -0.115 b, PA



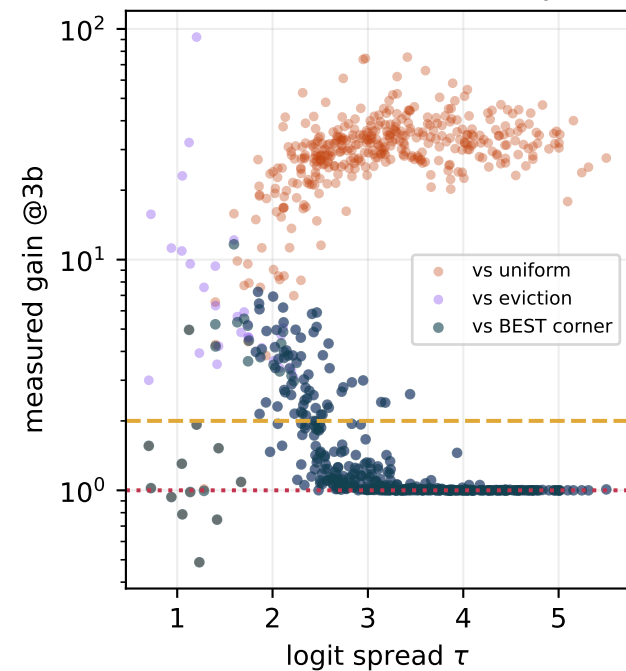
HEADS IN BAND: 20%



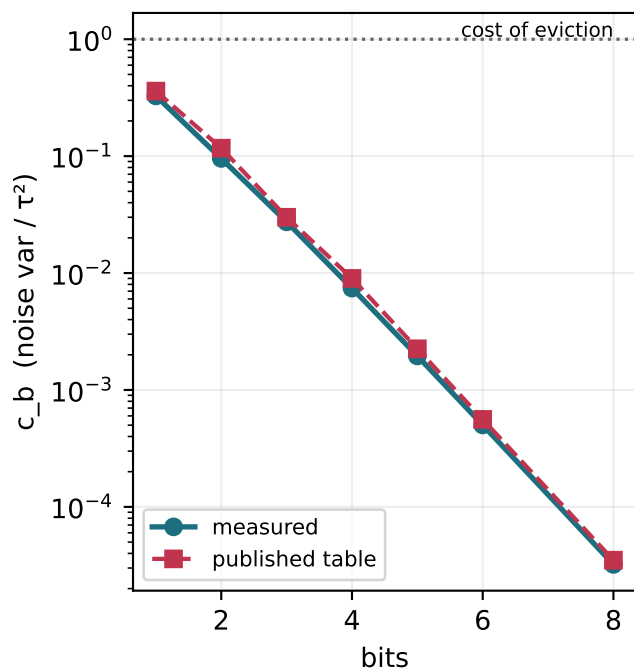
Which heads the router sends to SIEVE



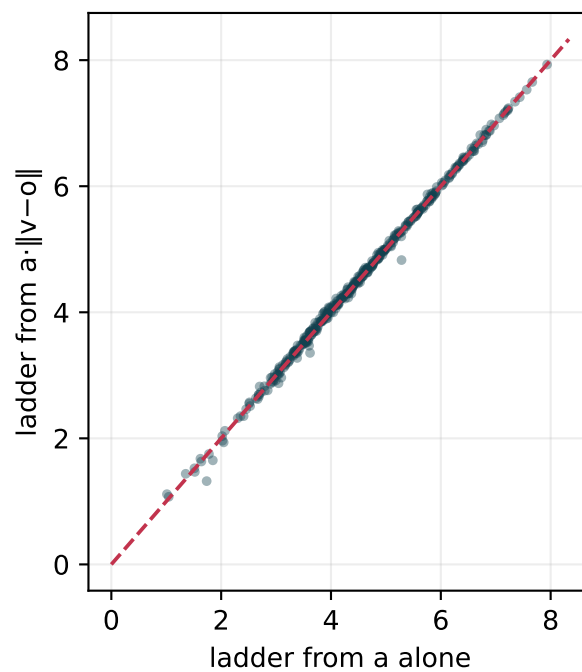
Where does the interior actually win?



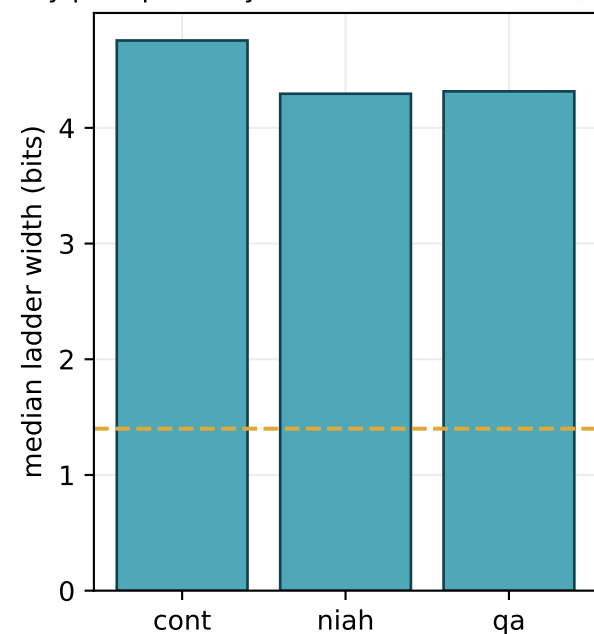
Measured vs assumed noise



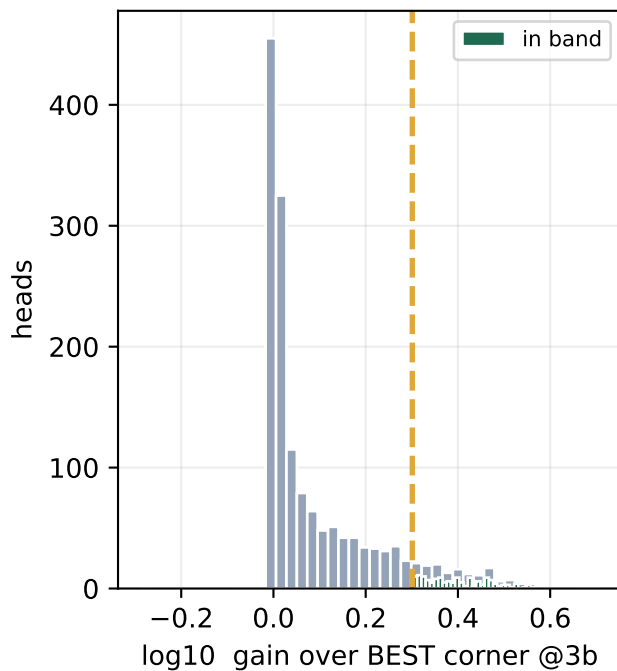
Does the value term matter?



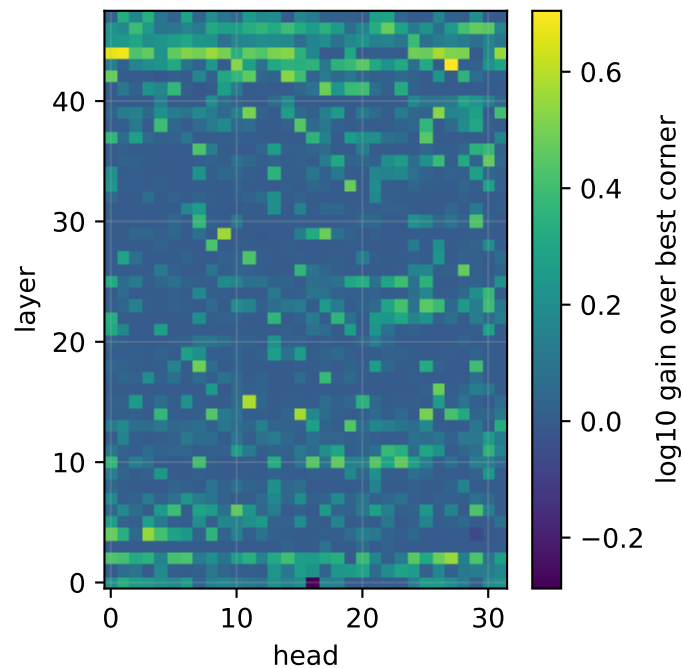
By prompt family — niah—cont -0.268 b, PA



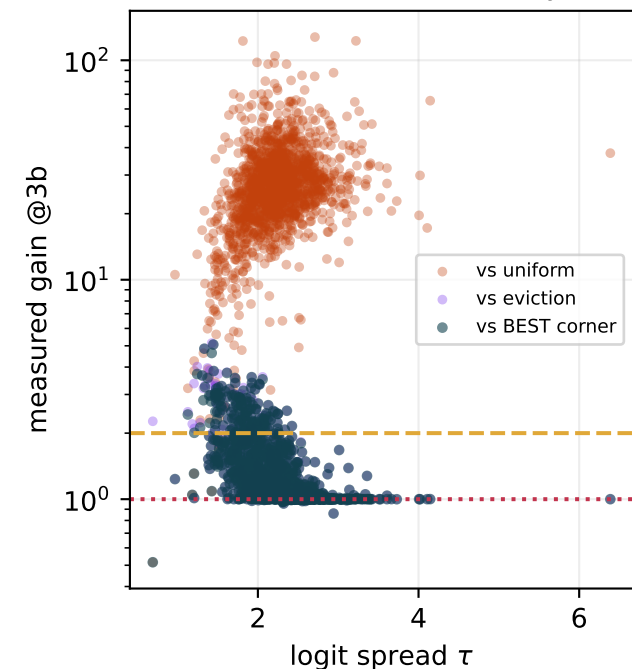
HEADS IN BAND: 10%



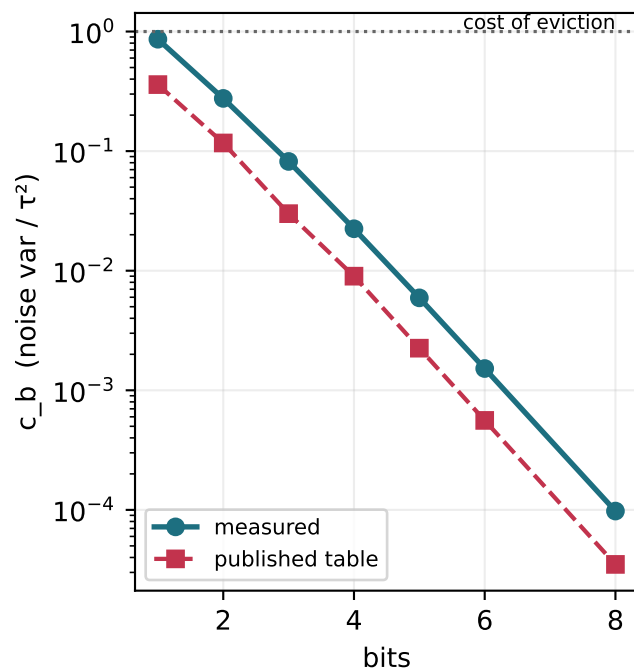
Which heads the router sends to SIEVE



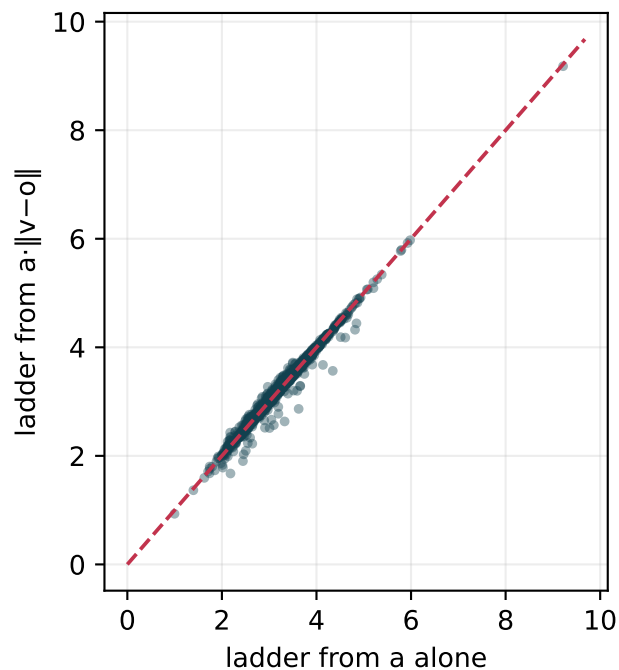
Where does the interior actually win?



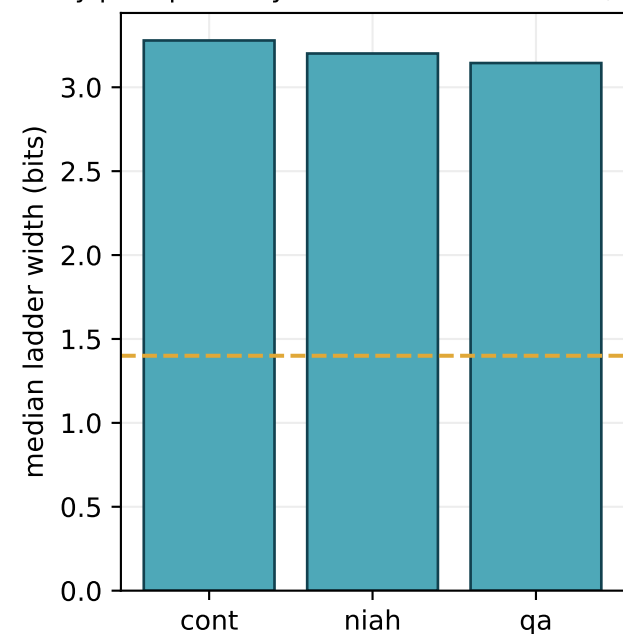
Measured vs assumed noise



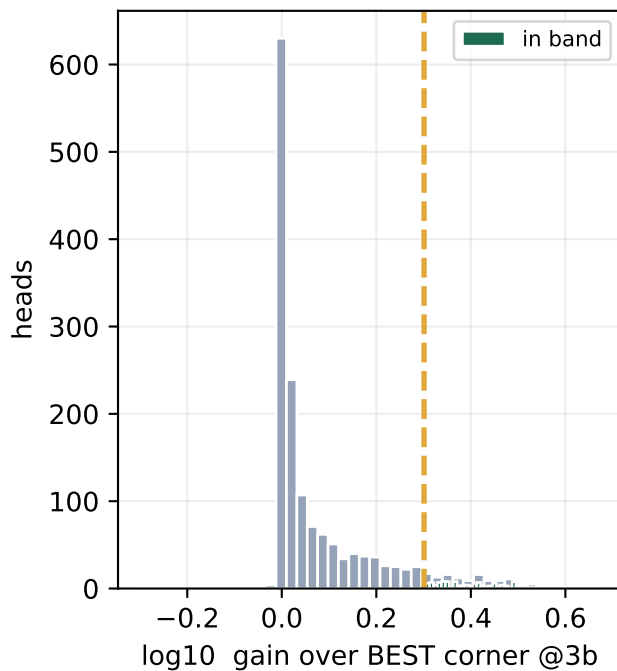
Does the value term matter?



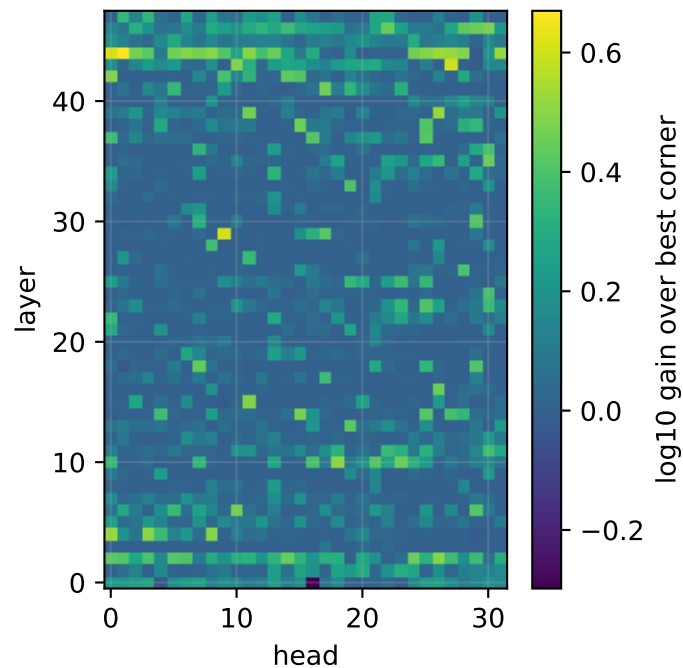
By prompt family — niah—cont -0.046 b, PA



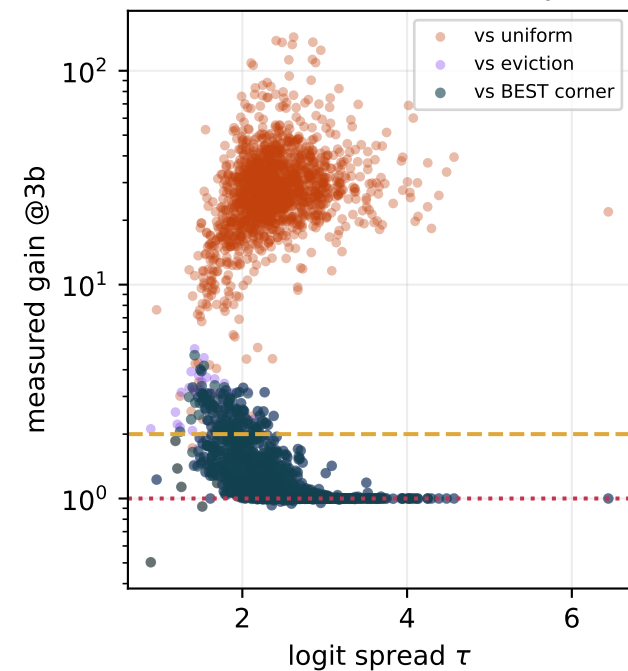
HEADS IN BAND: 8%



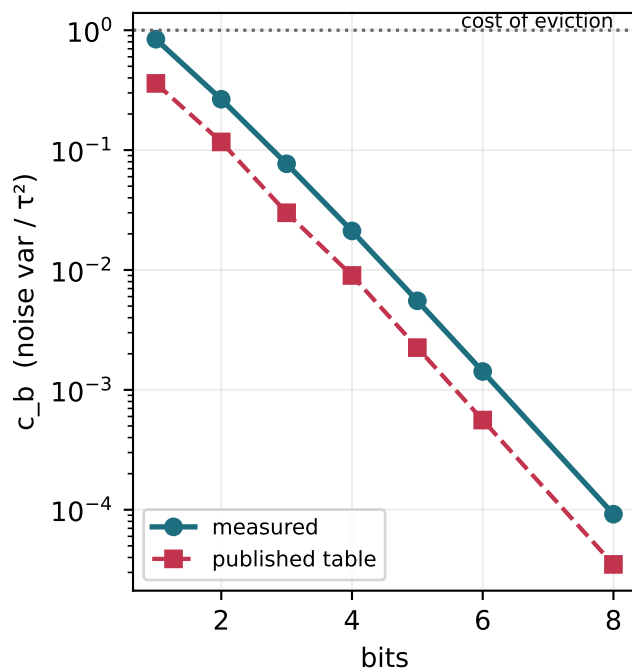
Which heads the router sends to SIEVE



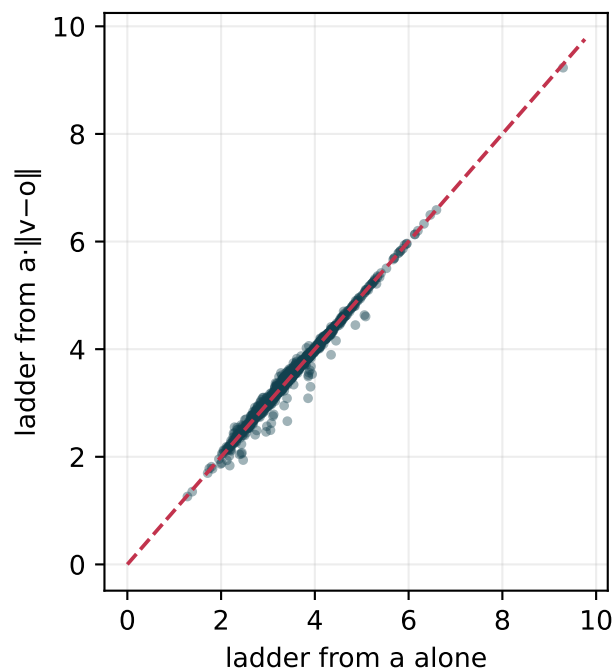
Where does the interior actually win?



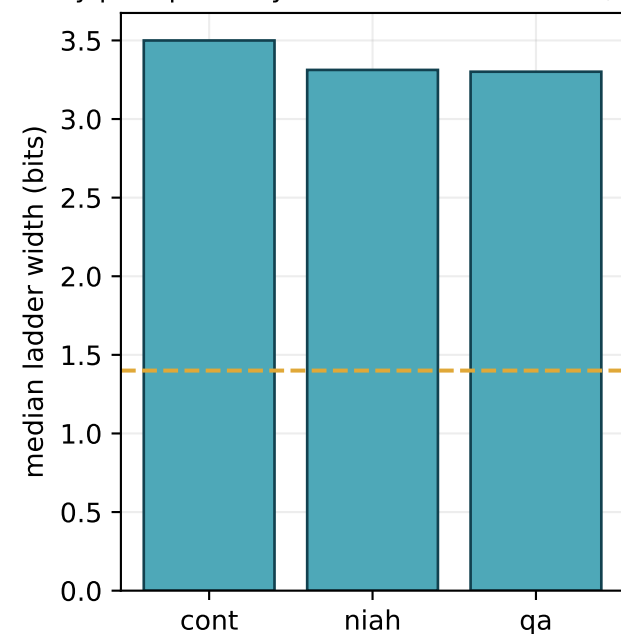
Measured vs assumed noise



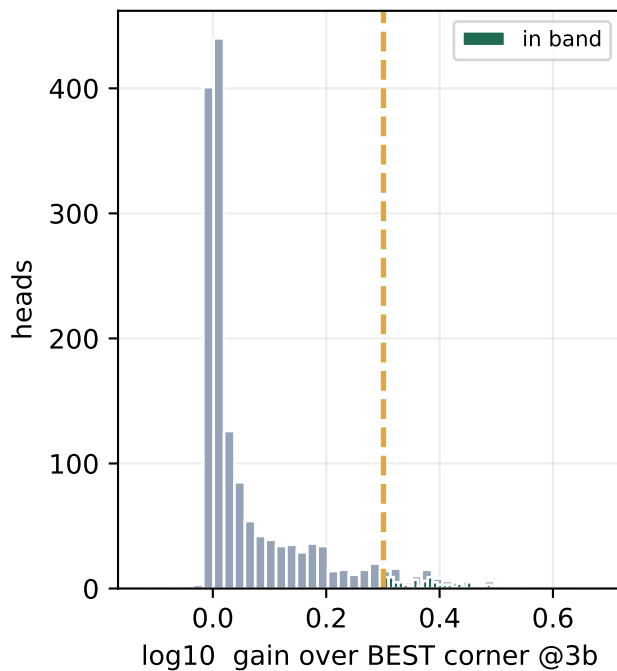
Does the value term matter?



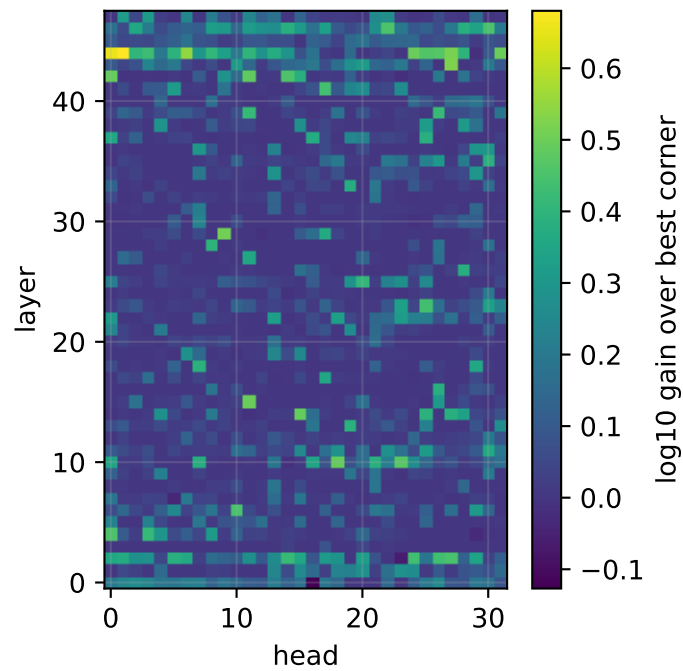
By prompt family — niah—cont -0.166 b, PA



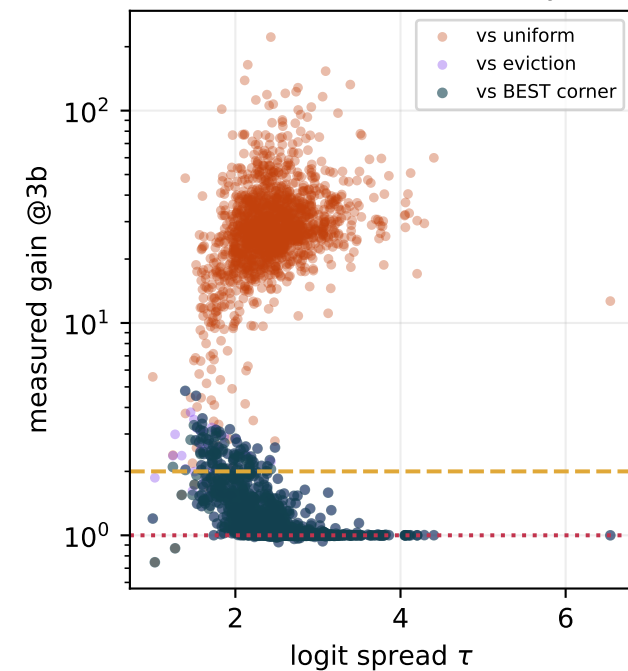
HEADS IN BAND: 6%



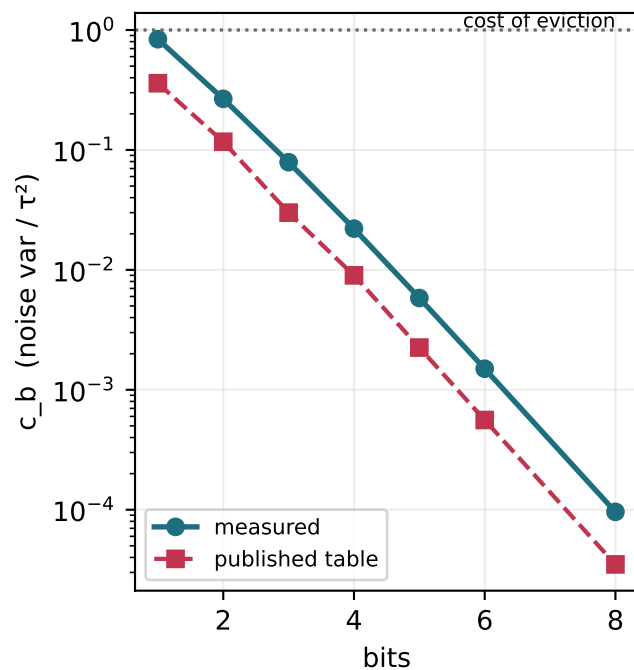
Which heads the router sends to SIEVE



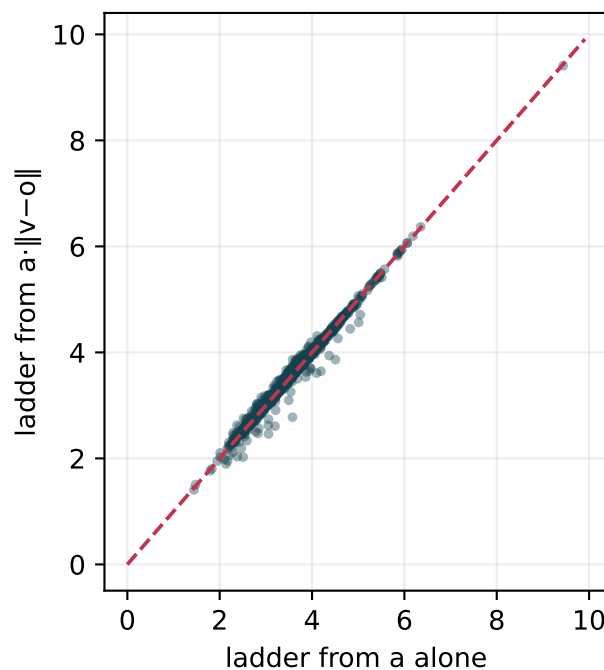
Where does the interior actually win?



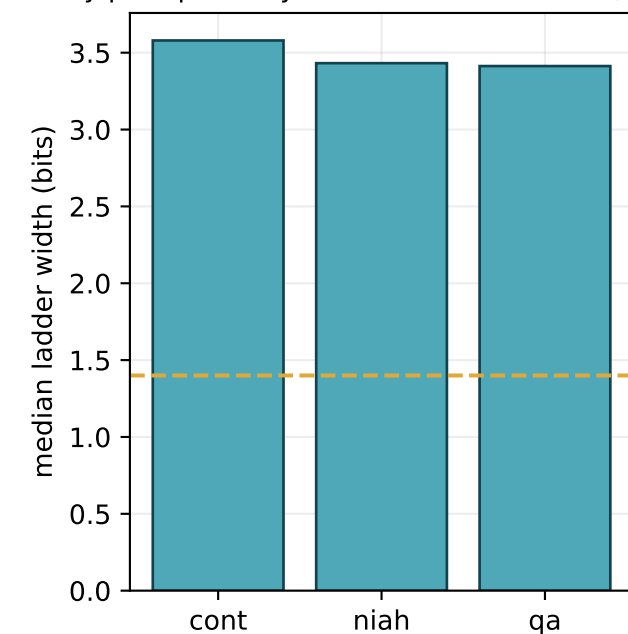
Measured vs assumed noise



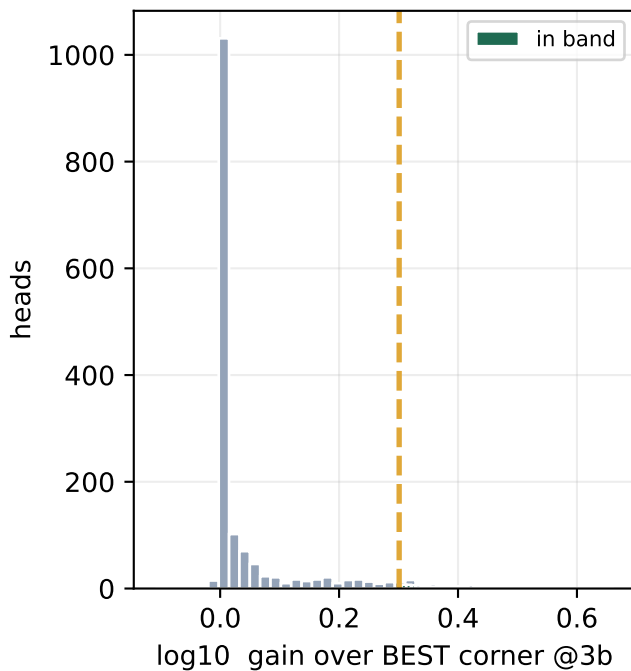
Does the value term matter?



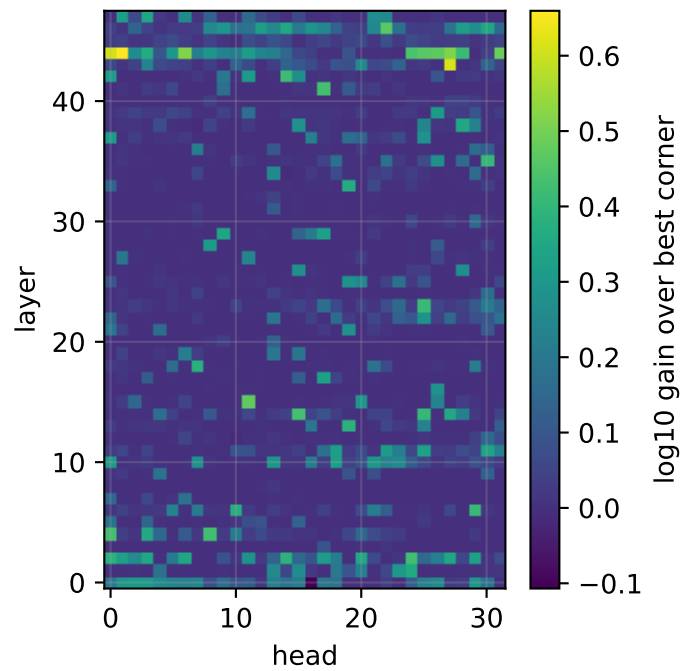
By prompt family — niah—cont -0.100 b, PA



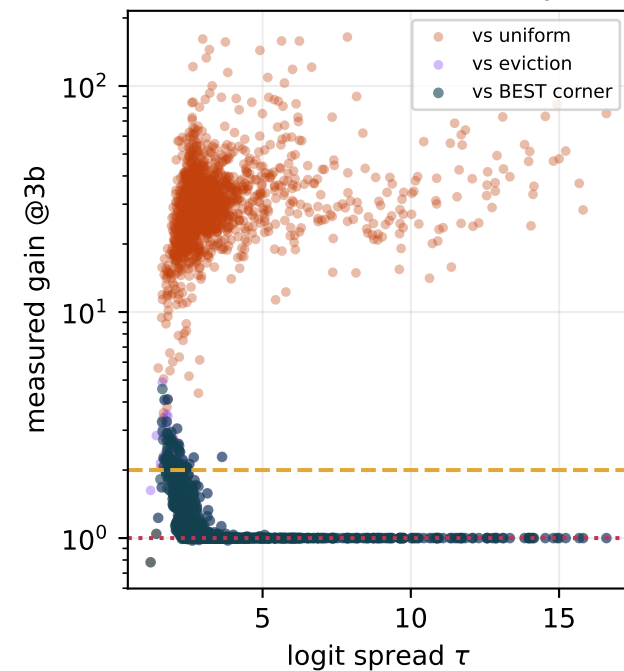
HEADS IN BAND: 4%



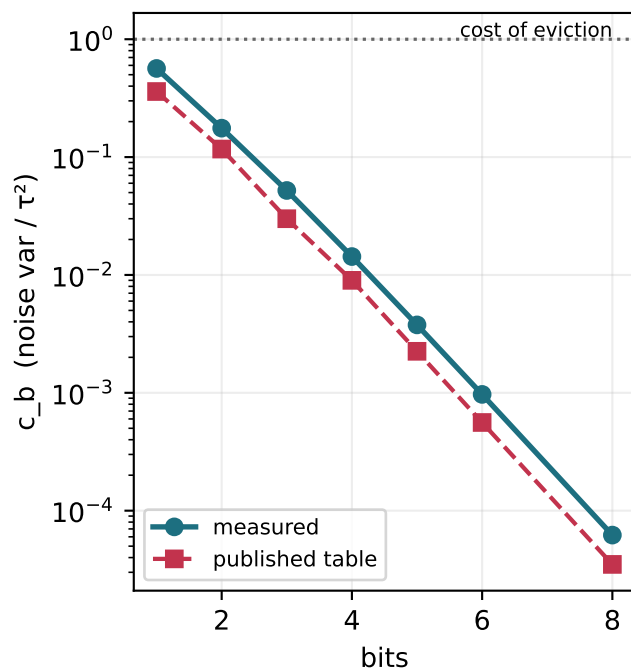
Which heads the router sends to SIEVE



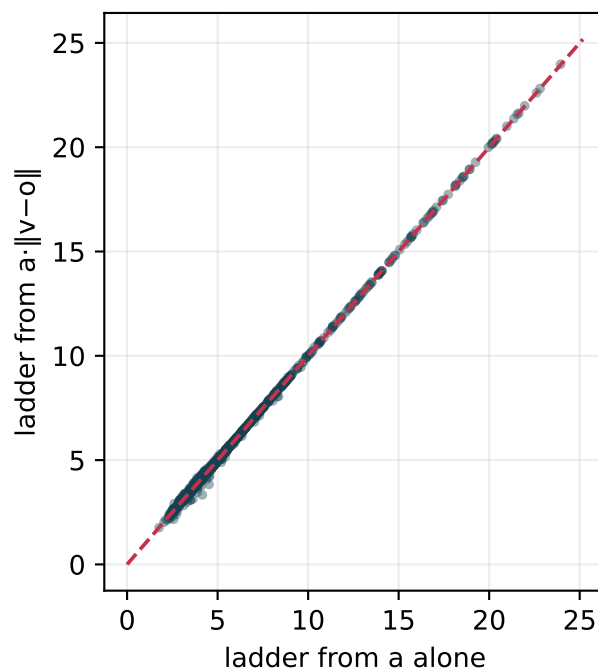
Where does the interior actually win?



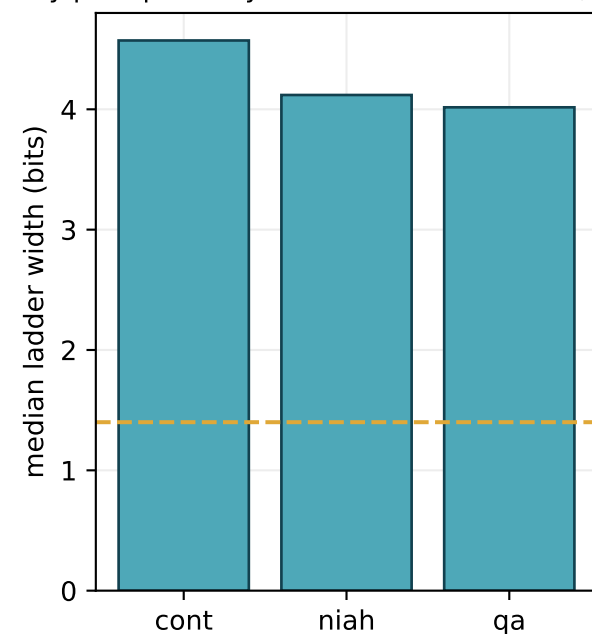
Measured vs assumed noise



Does the value term matter?

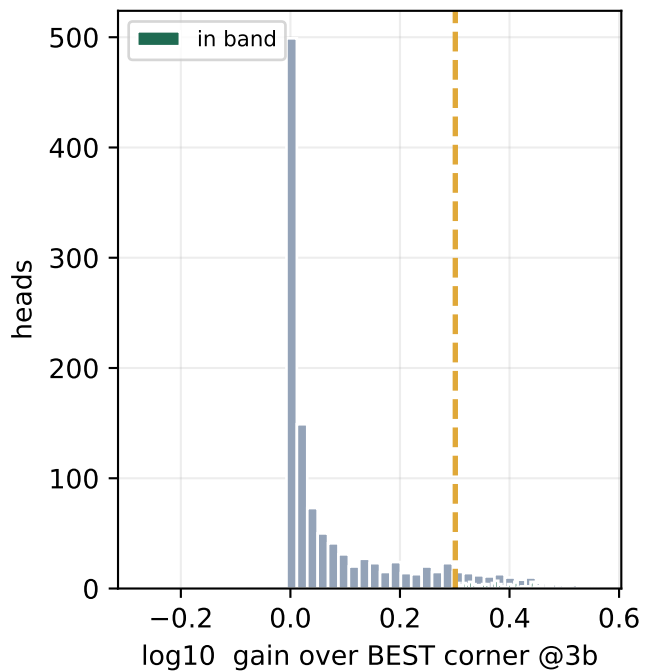


By prompt family — niah—cont -0.382 b, PA

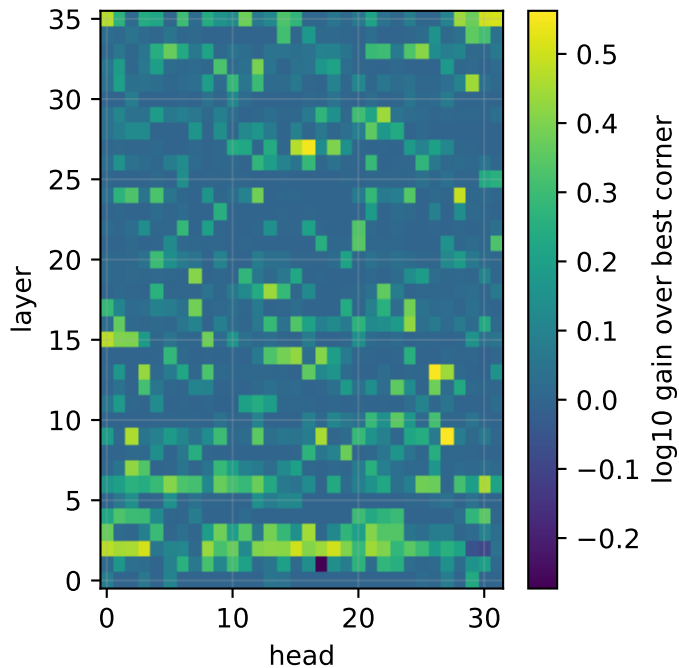


qwen3-8b — ctx 8,192

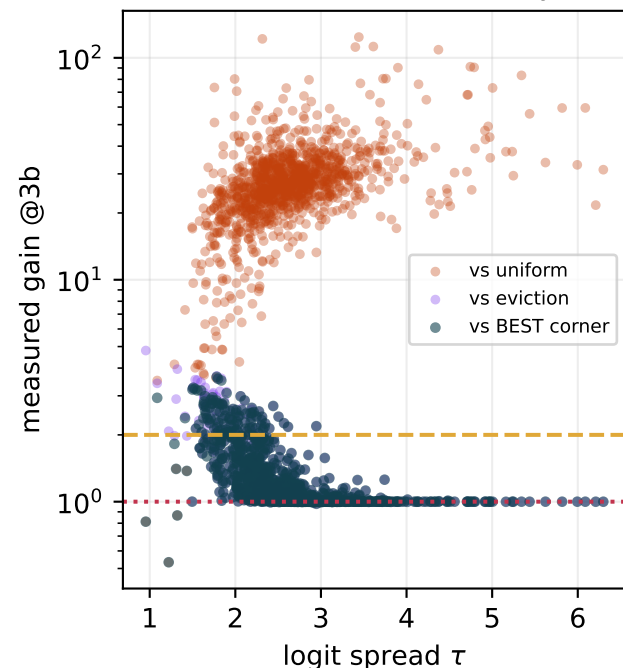
HEADS IN BAND: 9%



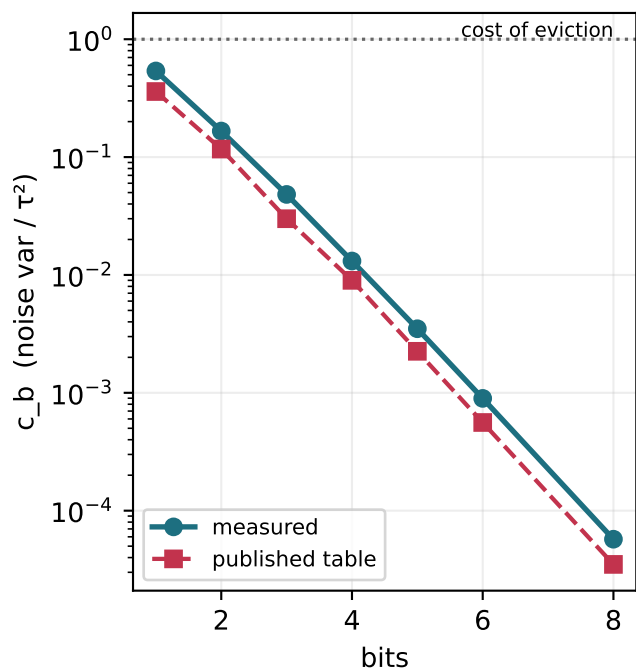
Which heads the router sends to SIEVE



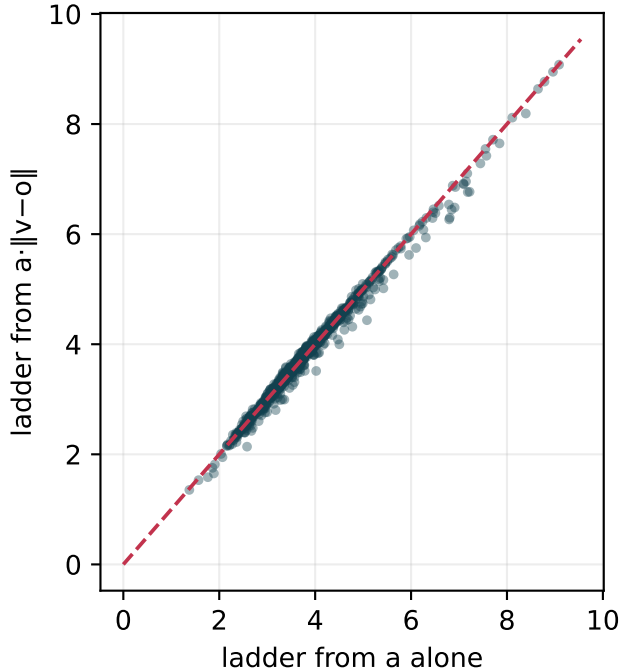
Where does the interior actually win?



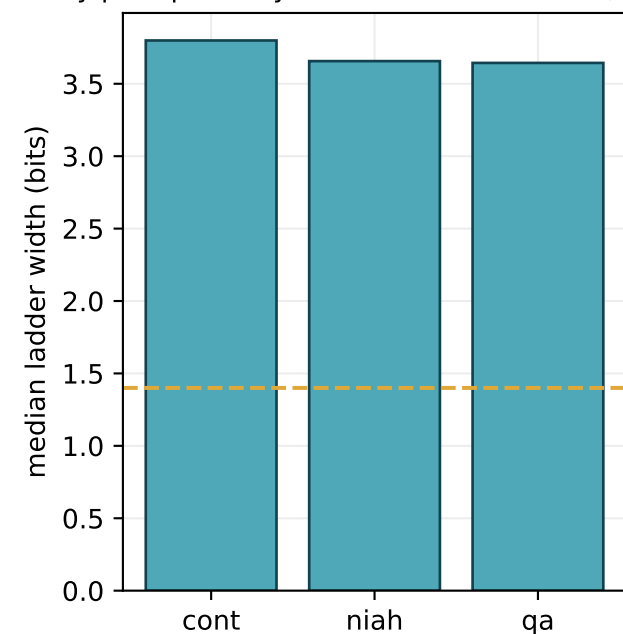
Measured vs assumed noise



Does the value term matter?

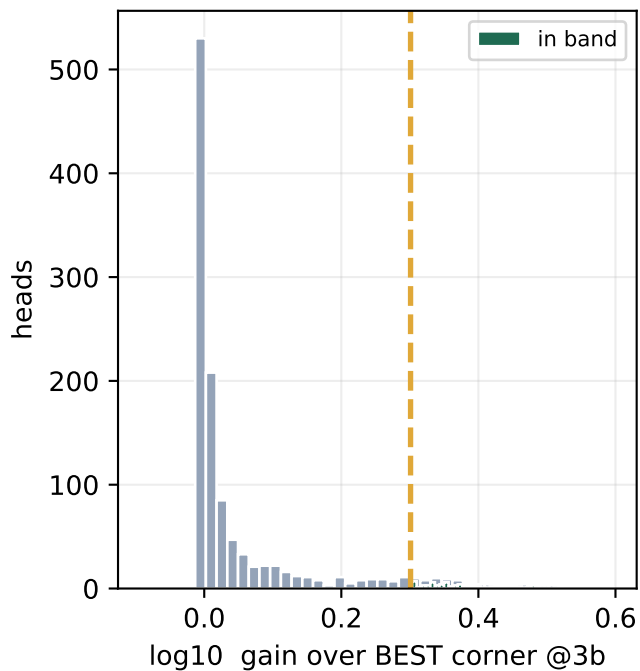


By prompt family — niah—cont -0.153 b, PA

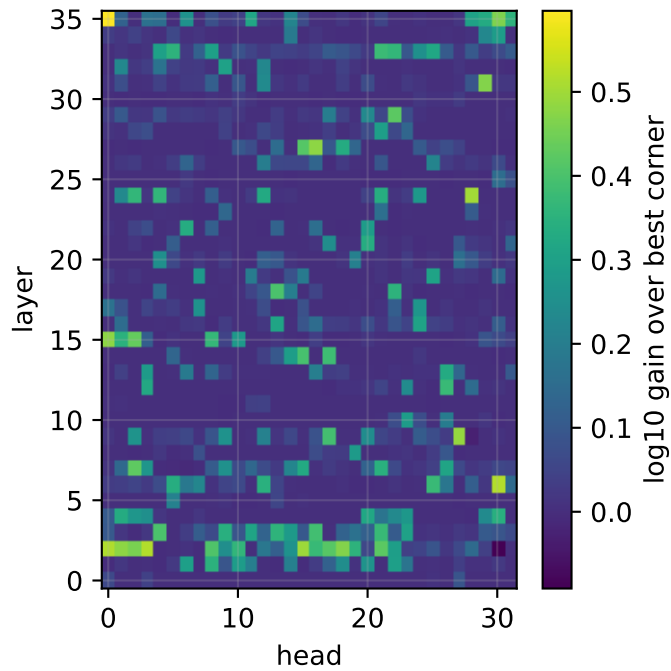


qwen3-8b — ctx 32,768

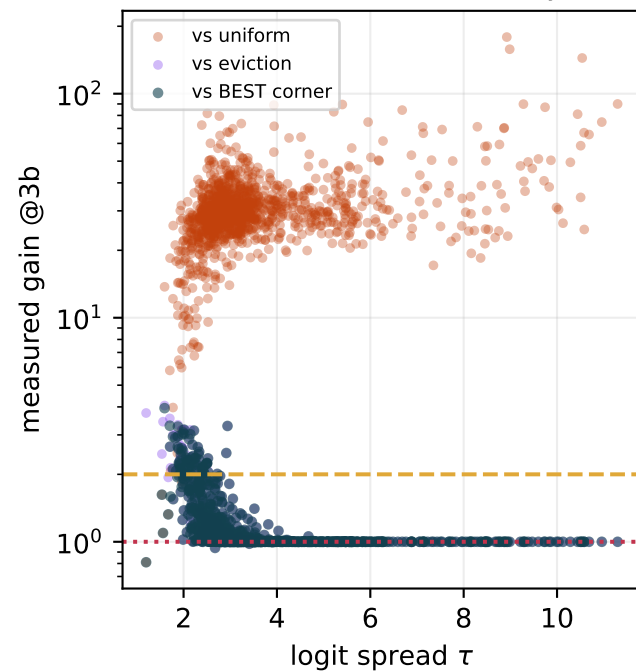
HEADS IN BAND: 6%



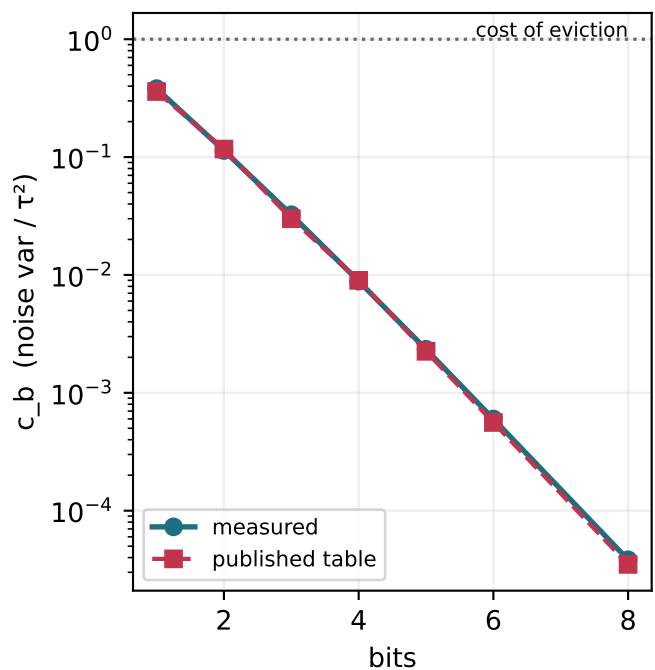
Which heads the router sends to SIEVE



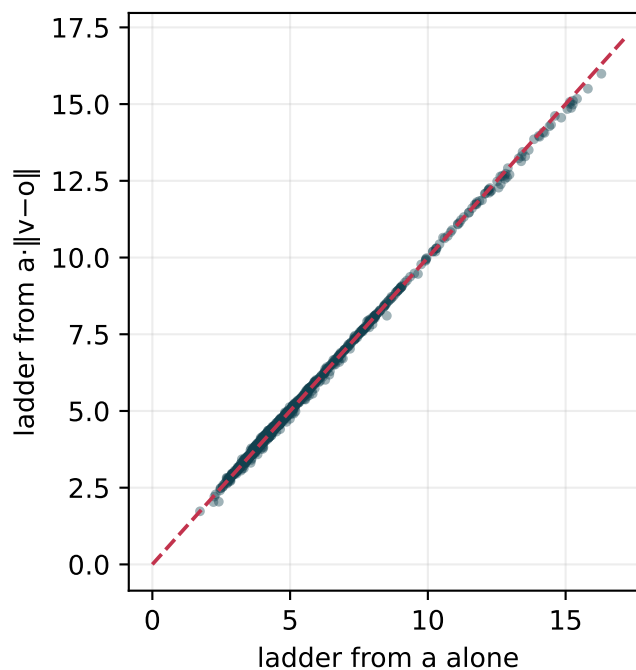
Where does the interior actually win?



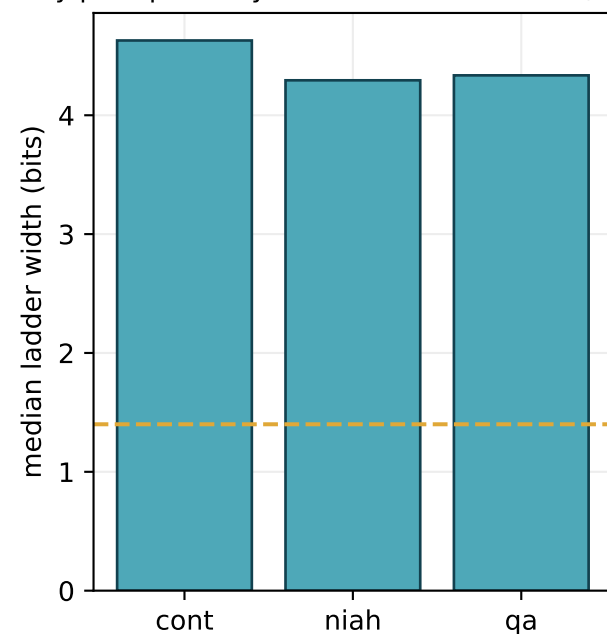
Measured vs assumed noise



Does the value term matter?

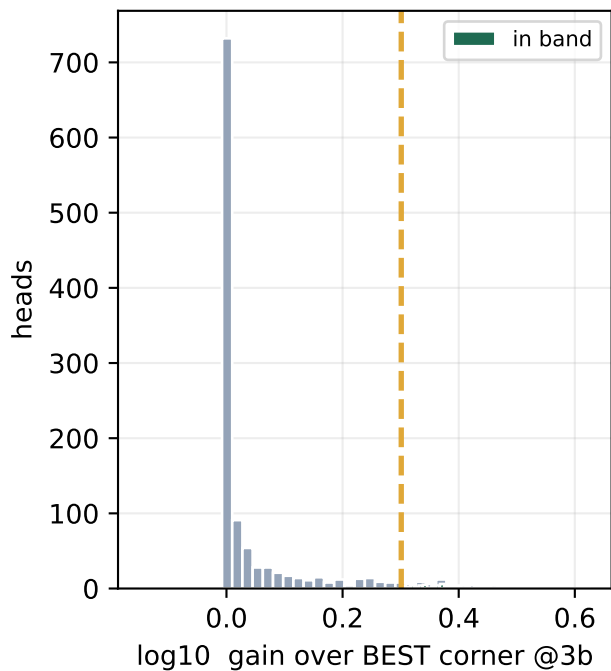


By prompt family — niah—cont -0.310 b, PA

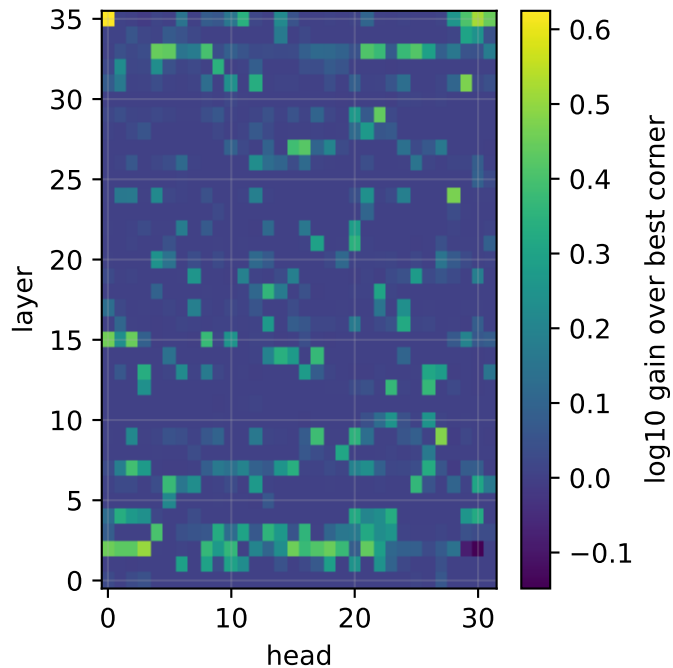


qwen3-8b — ctx 40,960

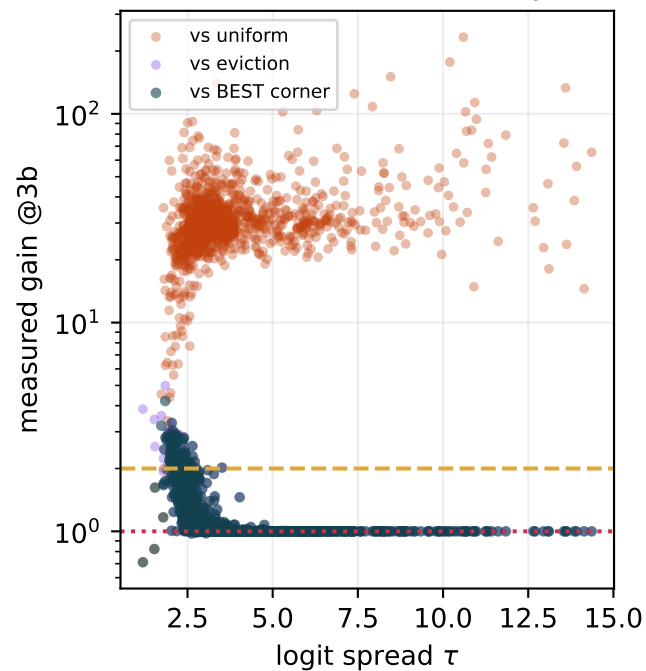
HEADS IN BAND: 6%



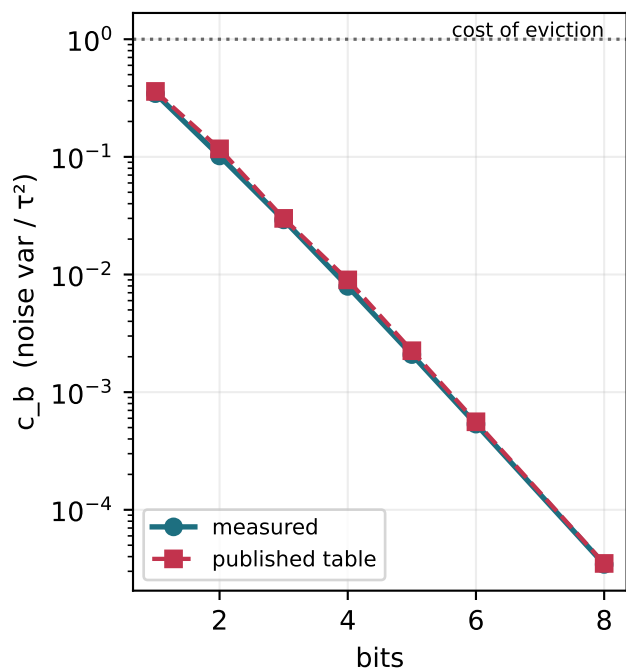
Which heads the router sends to SIEVE



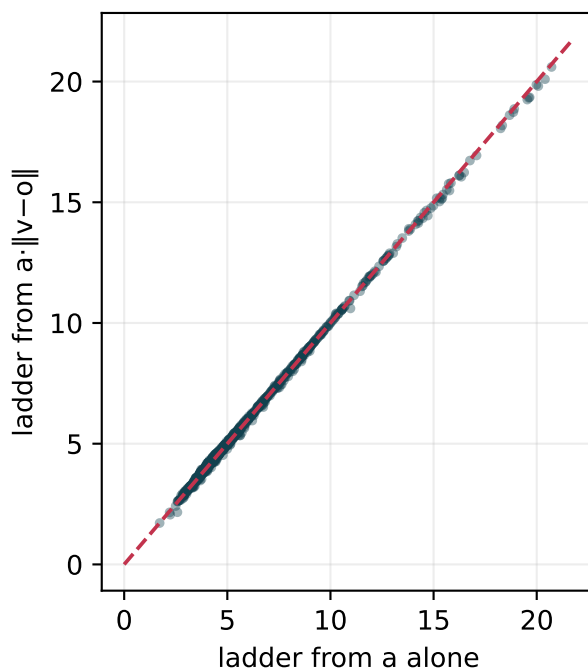
Where does the interior actually win?



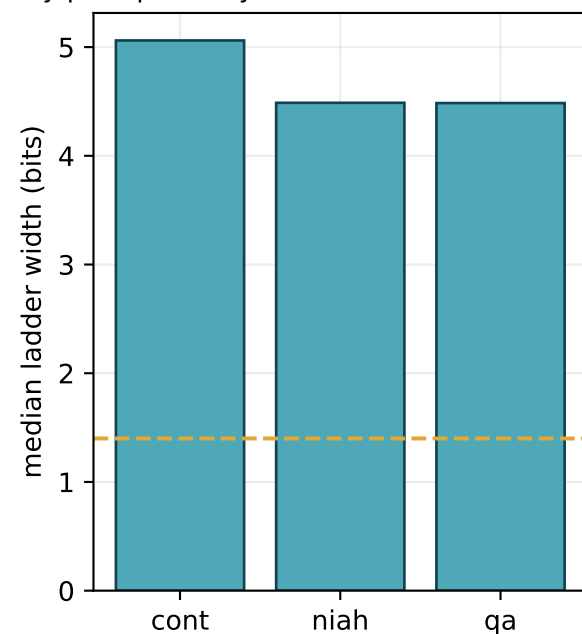
Measured vs assumed noise



Does the value term matter?



By prompt family — niah—cont -0.432 b, PAS



Band fraction vs context length (the ctx sweep)

